

Refinement of dose–effect models will continue in at least three areas, all of which will ultimately lead to more precise risk estimates. The first is better epidemiologic data—that is, epidemiologic studies with less uncertainty in each data point, thereby restricting the range of polynomials that fit the data. The second is extension of data into lower doses. The third is a greater understanding of fundamental mechanisms of radiation injury.

FACTORS THAT INFLUENCE DOSE-EFFECT MODELS

The effects of radiation are known to be modified by a number of factors. These effects can be expressed independently as a function of each of the variables. For example, age of the individual at the time of exposure is known to be important in estimating cancer risk. Furthermore, the time that has elapsed since a single exposure also influences the probability of occurrence of an effect. Table 27-2 gives the excess cancer deaths in a population of 100,000 exposed to 0.1 Sv (10 rem) according to the age at which the exposure occurred. For two reasons, the risk of cancer is greater when exposure occurs at an earlier age. First, if radiation exposure occurs at an earlier age, more years of life remain in which cancer may occur. Therefore, the lifetime risk of cancer is greater. Second, for most types of cancer the probability of cancer induction is actually greater on a yearly basis for exposures that occur at an earlier age.^{57–58}

Regardless of the age at irradiation, a latency period, an interval of time that must elapse before the rate of occurrence of a radiation-induced effect rises above the natural rate of occurrence of that disorder in a population, always occurs before appearance of a stochastic radiation-induced effect. The latency period varies with different disorders. For radiation-induced leukemia, the latency period may be as short as 2 years, whereas for solid tumors it may be more than 20 years and may vary

The NCRP recommends that astronauts receive no more than 0.5 Sv per year and no more than 0.25 Sv per month. The limit for exposure during a career as an astronaut depends upon the age of first exposure and gender. It varies from 1.0 Sv for females first exposed at age 25 to 4.0 Sv for males first exposed at age 55. The extra lifetime risk of fatal cancer for the maximum dose limit is $4.0 \text{ Sv} \times 4\% = 16\%$, meaning that the total lifetime risk of cancer for this astronaut is, theoretically, twice the typical risk of dying of cancer from all causes, which for a typical U.S. citizen is 16%.

A typical adult contains:
a total of ~9000 Bq of radioactivity
This includes:

4000 Bq of ^{14}C
3750 Bq of ^{40}K
600 Bq of ^3H
2 Bq of ^{226}Ra

and smaller amounts of ^{87}Rb , ^{238}U ,
 $^{228,230,232}\text{Th}$, $^{220,222}\text{Rn}$, ^{210}Pb , and ^{210}Po .

TABLE 27-2 Cancer Excess Mortality by Age at Exposure and Site for 100,000 Persons of Each Age Exposed to 0.1 Sv (10 rem)

Age at Exposure (yr)	Total	Leukemia	Nonleukemia	Breast	Respiratory	Digestive	Other
Males							
5	1276	111	1165		17	361	787
15	1144	109	1035		54	369	612
25	921	36	885		124	389	372
35	566	62	504		243	28	233
45	600	108	492		353	22	117
55	616	166	450		393	15	42
65	481	191	290		272	11	7
75	258	165	93		90	5	—
85	110	96	14		17	—	—
Average ^a	770	110	660		190	170	300
Females							
5	1532	75	1457	129	48	655	625
15	1566	72	1494	295	70	653	476
25	1178	29	1149	52	125	679	293
35	557	46	511	43	208	73	187
45	541	73	468	20	277	71	100
55	505	117	388	6	273	64	45
65	386	146	240	—	172	52	16
75	227	127	100	—	72	26	3
85	90	73	17	—	15	4	—
Average	810	80	730	70	150	290	220

^aAverages are weighted for the age distribution in a stationary population having U.S. mortality rates and have been rounded to the nearest 10.

Source: Committee on the Biological Effects of Ionizing Radiations. *Health Effects of Exposure to Low Levels of Ionizing Radiation*, Report No.5. Washington, D.C., National Academy Press, 1990, p. 175. Used with permission.

You eat approximately 40 Bq of radioactivity per year.

Because of radioactivity present naturally in your body, you give yourself a dose equivalent of about 0.50 mSv (50 mrem) each year to your bone marrow, 1.9 mSv (190 mrem) to your bone cells (osteocytes), and 0.35 mSv (35 mrem) to your soft tissues [the total effective dose equivalent is 0.4 mSv (40 mrem)].

with the age of the individual at the time of exposure.⁵⁹ The latency period is also probabilistic, with values quoted for given disorders representing population averages.

There is some disagreement about the relationship between radiogenic stochastic effects and the natural incidence of the effects as a function of time after exposure. Two models have been proposed to describe this relationship. The absolute-risk model assumes that a constant increase in incidence of the effect takes place at all times after the latency period. The relative-risk model assumes that the radiation-induced incidence of an effect varies with the natural incidence of the effect at any time after the latency period. Thus, as natural incidence increases with age for many cancers, radiogenic cancer incidence increases also. Both models allow for a smooth increase in incidence of the effect during the latency period.

One of the current challenges of epidemiology is to determine which model best fits the available data. At the present time, both models fit the data reasonably well. Other factors being equal, the relative-risk model tends to predict a greater lifetime risk of cancer, because the incidence of disorders such as cancer increases with age. That is, the relative-risk model predicts that the number of cancers appearing later in life is greater than would be estimated from the absolute-risk model. Data concerning the largest and most exhaustively studied population, the atomic bomb survivors, were reviewed by the US National Research Council after 46 years (1945 to 1990) of study. Both absolute- and relative-risk models fit the data for different cancers in that population. However, the relative-risk model appears to yield more accurate estimates for most cancers, and it is the only model used in the (BEIR V) report from the National Research Council.⁶⁰

■ ESTIMATING RISKS OF RADIATION: BEIR REPORT

A number of scientific and advisory groups have published risk estimates for ionizing radiation. According to current NCRP and ICRP risk estimates, the risk of fatal cancer is 4% per Sv (4.0×10^{-4} per rem). For radiographers, radiologists, and medical physicists, the annual dose limit is 0.05 Sv (50 mSv, 5 rem) and the cumulative limit is 10 mSv times the age in years. Therefore, a 55-year-old radiographer should have received a total occupational exposure of no more than 550 mSv (0.55 Sv, 55 rem). This maximum level of exposure is associated with an estimated $0.55 \times 4\% = 2.2\%$ increase in risk of fatal cancer.

One of the most recent and widely quoted reports on risk estimation is Report No. 5 of the National Academy of Sciences Committee on the Biological Effects of Ionizing Radiation (BEIR V). Table 27-3 summarizes the risk estimates for cancer from that publication.

The BEIR V report uses the linear dose–effect model for all cancers other than leukemia, along with the linear quadratic model for leukemia. The effects of age at irradiation and time elapsed since irradiation are considered with latency period beginning 2 years after irradiation for leukemia and 10 years after irradiation for other cancers. In keeping with recent trends in stochastic effects among atom bomb survivors, the relative-risk model of cancer incidence as a function of age is used.

The BEIR V report estimates (see Table 27-3) that if a population of 100,000 received a single exposure to all body organs of 0.1 Sv (10 rem), then there would be an extra (i.e., radiation-induced or radiogenic excess) 770 cases of cancer if the population were male and 810 if the population were female. Given the 90% confidence limits for these estimates (540 to 1240 for males, 630 to 1160 for females), it is reasonable to adopt an average of 800 cancers for a population of 100,000 with a reasonable distribution of males and females.

Many factors enter into the average estimate of radiation risk. The estimate of 800 excess cases per 100,000 per 0.1 Sv is the result of a calculation in which excess fatal cancers were estimated for single exposures occurring at different ages of individuals. The individuals are members of a hypothetical population having the same age distribution as the U.S. population. The results of these calculations are then averaged to arrive at the figure 800.

TABLE 27-3 Excess Cancer Mortality Estimates and Their Statistical Uncertainty: Lifetime Risks per 100,000 Exposed Persons^a

	Male			Female		
	Total	Nonleukemia ^b	Leukemia ^c	Total	Nonleukemia	Leukemia
Single exposure to 0.1 Sv (10 rem)	770	660	110	810	730	80
90% confidence limits	540–1240	420–1040	50–280	630–1160	550–1020	30–190
Normal expectation	20,510	19,750	760	16,150	15,540	610
% of normal	3.7	3.3	15	5	4.7	14
Total years of life lost	12,000			14,500		
Average years of life lost per excess death	16			18		
Continuous lifetime exposure ^d to 1 mSv/yr (0.1 rem/yr)	520	450	70	600	540	60
90% confidence limits	410–980	320–830	20–260	500–930	430–800	20–200
Normal expectation	20,560	19,760	790	17,520	16,850	660
% of normal	2.5	2.3	8.9	3.4	3.2	8.6
Total years of life lost	8100			10,500		
Average years of life lost per excess death	16			18		
Continuous exposure ^d to 0.01 Sv/yr (1 rem/yr) from age 18 until age 65	2880	2480	400	3070	2760	310
90% confidence limits	2150–5460	1670–4560	130–1160	2510–4580	2120–4190	110–910
Normal expectation	20,910	20,140	780	17,710	17,050	650
% of normal	14	12	52	17	16	48
Total years of life lost	42,200			51,600		
Average years of life lost per excess death	15			17		

^aBased on an equal dose to all organs and the committee's preferred risk models—estimates are rounded to nearest 10.

^bSum of respiratory, breast, digestive, and other cancers.

^cEstimates for leukemia contain an implicit dose rate reduction factor.

^dA dose rate reduction factor has not been applied to the risk estimates for solid cancers.

Source: Committee on the Biological Effects of Ionizing Radiation. *Health Effects of Exposure to Low Levels of Ionizing Radiation*, Report No. 5, Washington, D.C., National Academy Press, 1990, p. 172. Used with permission.

Another approach to risk estimation is the specification of risk to an individual who has been irradiated. One could state that the risk to an individual who has received 0.1 Sv (10 rem) in a single whole-body exposure is approximately 800 chances per 100,000 of acquiring a fatal radiation-induced cancer. This number, which may also be expressed as 8 chances in 1000, 8×10^{-3} , or 0.8%, applies only to a 0.1-Sv exposure. For higher exposures, the linear dose–effect model allows increasing the risk in proportion to the dose (e.g., doubling the risk as the dose is doubled). This is appropriate up to the dose level where incidence levels off, which is beyond dose levels for which data have been obtained from epidemiologic studies. In the BEIR V committee, actual data above 4 Sv (400 rem) are not used.⁶¹ The BEIR committee's model, if extrapolated to larger doses, corresponds to an 8% risk of fatal cancer per sievert. This value is greater than the NCRP and ICRP estimate of 4% per sievert, although the confidence limits of these estimates (Table 27-3) overlap. Many epidemiologists and actuarial specialists summarize the various estimates with an easily remembered “rule of thumb” of “5% increase in fatal cancer per sievert.”

One may also scale or extrapolate the risk estimates downward in proportion to the dose. However, caution should be used for the reasons cited earlier. Dose levels in the epidemiologic studies on which risk estimates are based are above 0.1 Sv. Using the linear dose–effect model to extrapolate to lower dose levels should be considered to be simply an estimate, quite possibly an overestimate, of risk.

The natural incidence of fatal cancers is also given in Table 27-3. Out of a population of 100,000 typical U.S. citizens followed until death has occurred, it is expected that 16,000 to 20,000 of the deaths would be due to cancer. Thus 0.1 Sv (10 rem)

to a population would be estimated to increase the number of fatal cancers by about 4%. Average years of life lost per radiogenic cancer and total years of life lost in the population are also shown.

Example 27-1

Find the risk of contracting a fatal cancer from a single, whole-body dose of 0.02 Sv (2 rem), and compare this risk with the natural incidence of fatal cancer.

By extrapolation from Table 27-3, since 0.02 Sv is 20% of 0.1 Sv, the risk is

$$\frac{0.2(770)}{100,000} = 1.54 \times 10^{-3} \text{ for males}$$

and

$$\frac{(0.2)(810)}{100,000} = 1.62 \times 10^{-3} \text{ for females}$$

This represents a fraction of the naturally occurring cancers in a population as follows:

$$\frac{(0.2)(770)}{20,510} = 7.5 \times 10^{-3} = 0.75\% \text{ for males}$$

and

$$\frac{0.2(810)}{16,150} = 10.0 \times 10^{-3} = 1\% \text{ for females}$$

For several years before BEIR V, a range of 1×10^{-4} to 2×10^{-4} per rem (0.01 Sv) was widely used as an overall estimate of radiation risk. There are several reasons for the increase to the currently accepted value of 8×10^{-3} per 0.1 Sv. One reason is that as more years of follow-up have been accumulated in groups such as the atom bomb survivors, it has been found that the number of tumors, particularly solid tumors, has increased beyond earlier predictions. This finding is in keeping with the predictions of the relative-risk model.

A second reason for the increase is that dose estimates for atom bomb survivors have been redone^{62,63} from information that had previously been classified. It is now thought that the γ -ray doses in both Hiroshima and Nagasaki were lower than previously believed and that the neutron component of the dose was much less significant. Therefore, the radiogenic cancers seen in those populations may have been caused by smaller doses than had been previously assumed.

The BEIR V report also gives estimates of the risk of the continuous exposure. Table 27-3 shows risks calculated for (a) a continuous lifetime exposure to all body organs of 1 mSv/yr (0.1 rem/yr) and (b) a continuous exposure to all body organs of 10 mSv/yr (1 rem/yr) delivered between ages 18 and 65. The former simulates an upper level for exposure of the general public, and the latter simulates an upper level for occupational exposure. These estimates are obtained by summing the age-specific incidence data for doses delivered over time. They indicate that continuous lifetime exposure yields a risk of fatal cancer that is 2% to 3% of the normal incidence and that occupational exposure yields a risk that is 14% to 17% of the normal incidence. However, these estimates should be interpreted as overestimates because the BEIR V committee chose not to apply a dose rate effectiveness factor (DREF) to the data. As was discussed earlier, the inclusion of a DREF would be expected to lower risk estimates by a factor of 2 to 3.⁶⁴

The BEIR V report also contains an extensive compilation of organ-specific models for cancer risk. These data are more appropriate for computing risk to a patient in radiology than is the whole-body risk estimate of 800 cases of fatal cancer per 100,000 for each 0.1 Sv, because almost all diagnostic studies involve only partial-body exposure. As with whole-body irradiation, caution should be used in applying risk estimates to dose levels below those for which epidemiologic data exist.

The National Academy of Sciences is revising the BEIRV report on the Health Effects of Exposure to Low Levels of Ionizing Radiations. As with previous reports, panels of experts are reviewing the worldwide literature on the subject and are conducting their own analyses of various predictive models. The new report, is expected to include new information on:

- Mechanisms of the cellular and molecular events involved in the neoplastic process
- The role of genetics on radiation susceptibility as revealed in recent studies involving genetically engineered mice
- New evidence for specific molecular events that affect the shape of the dose–effect curve at low doses. These effects include:
 - DNA repair
 - Signal transduction
 - Chromosomal instability
 - Genetic adaptation
- An assessment of the current status of all models of carcinogenesis, including the possibility of thresholds, adaptive responses, and hormetic effects.

Effects on Embryo

Irradiation of the embryo is estimated by the BEIR V Committee to increase the incidence of mental retardation by 4% for a 0.1-Sv (10 rem) dose delivered during weeks 8 to 15 of gestation. As gestational age increases beyond week 15, the risk decreases. No statistically significant increase is noted for irradiation prior to week 8 or beyond week 25.

There are also data to support the hypothesis that irradiation of the embryo increases the risk of childhood (conception to the age of 15 years) cancer. Both leukemia and nervous system tumors have been linked to radiation exposure *in utero*. A risk of approximately 200 excess fatal cancers per 10,000 person-Gy is cited in BEIR V.

Genetic Effects

At the present time, there are no statistically significant human epidemiologic data from which information on the heritable effects of radiation can be derived. Risk estimates given by the BEIR V committee are based on chromosomal aberration data from humans coupled with studies of other mammals, principally mice. These estimates are summarized in Table 27-4. BEIR V estimates that the doubling dose for humans, the dose at which the natural mutation rate is doubled, is 1 Gy (100 rad). The risk of severe genetic damage appearing in the next few generations is 0.8% per Sv (0.8×10^{-4} per rem).

Human populations, fortunately, have not been exposed to high doses in sufficiently large numbers (the average dose to the atom bomb survivors was approximately 50 Gy) to provide a definitive demonstration of genetic risk. This provides evidence that, in all likelihood, the current estimates of genetic risk are conservative.

The National Academy of Sciences Report No. 6 of The Committee on Biological Effects of Ionizing Radiations (BEIR VI) was on the subject of the health effects of exposure to radon. The new report on the Health Effects of Exposure to Low Levels of Ionizing Radiations will be available from the National Academy Press (books.nap.edu) sometime after this text has gone to press.

Because of the radionuclides you contain within you body, if you sleep with someone (or somehow spend eight hours very close to someone each day) you give them (and they give you) an extra 0.001 mSv (0.1 mrem).

Mental retardation is a potential bioeffect of radiation exposure *in utero*. The risk of occurrence is greatest for exposures between the eighth and the fifteenth week of gestation. It is during this period that significant neurological organogenesis (differentiation of gray and white matter) occurs.

SOURCES OF INFORMATION

A number of scientific organizations have examined the biologic effects of ionizing radiation. These groups have reviewed the scientific literature and in many cases made risk estimates or recommendations concerning radiation safety. A listing is shown in Table 27-5 of organizations whose reports should be available in any major library or on the World Wide Web.

TABLE 27-4 Estimated Induced Mutation Rates per Rad (Primarily Mouse)

<i>Genetic End Point, Cell Stage, Sex</i>	<i>Low-LET Radiation Exposure</i>		
	<i>High Dose Rate</i>	<i>Low Dose Rate</i>	<i>Fission Neutrons (Any Dose Rate)</i>
Dominant lethal mutations			
Postgonial, male	10×10^{-4} /gamete	5×10^{-4} /gamete	75×10^{-4} /gamete
Gonial, male	10×10^{-5} /gamete	2×10^{-5} /gamete	40×10^{-5} /gamete
Recessive lethal mutations			
Gonial, male	1×10^{-4} /gamete		
Postgonial, female	1×10^{-4} /gamete		
Dominant visible mutations			
Gonial, male	2×10^{-5} /gamete		
Skeletal	5×10^{-7} /gamete		
Cataract	$5\text{--}10 \times 10^{-7}$ /gamete		
Other	$5\text{--}10 \times 10^{-7}$ /gamete	1×10^{-7} /gamete	25×10^{-7} /gamete
Postgonial, female	$5\text{--}10 \times 10^{-7}$ /gamete		
Recessive visible mutations (specific locus tests)			
Postgonial, male	65×10^{-8} /locus		
Postgonial, female	40×10^{-8} /locus	$1\text{--}3 \times 10^{-8}$ /locus	145×10^{-8} /locus
Gonial, male	22×10^{-8} /locus	7×10^{-8} /locus	125×10^{-8} /locus
Reciprocal translocations			
Gonial, male			
Mouse	$1\text{--}2 \times 10^{-4}$ /cell	$1\text{--}2 \times 10^{-5}$ /cell	$5\text{--}10 \times 10^{-4}$ /cell
Rhesus	2×10^{-4} /cell		
Marmoset	7×10^{-4} /cell		
Human	3×10^{-4} /cell		
Postgonial, female			
Mouse	$2\text{--}6 \times 10^{-4}$ /cell		
Heritable translocations			
Gonial, male	4×10^{-5} /gamete		
Postgonial, female	2×10^{-5} /gamete		
Congenital malformations			
Postgonial, female	2×10^{-4} /gamete		
Postgonial, male	4×10^{-5} /gamete		
Gonial, male	$2\text{--}6 \times 10^{-5}$ /gamete		
Aneuploidy (trisomy)			
Postgonial, female			
Preovulatory oocyte	6×10^{-4} /cell		
Less mature oocyte	6×10^{-5} /cell		

Source: *Committee on the Biological Effects of Ionizing Radiations: Health Effects of Exposure to Low Levels of Ionizing Radiation*, Report No. 5 (BEIR V), Washington, D.C., National Academy Press, 1990, p. 100. Used with permission.

TABLE 27-5 Reports and Recommendations of Scientific Committees on Biological Effects of Ionizing Radiation

UNSCEAR	United Nations Scientific Committee on the Effects of Atomic Radiation
BEIR	National Academy of Sciences—National Research Council, Committee on the Biological Effects of Ionizing Radiation, Washington, DC
NIH	National Institutes of Health Ad Hoc Working Group to Develop Radioepidemiological Tables; reports available from U.S. Government Printing Office
ICRP	International Commission on Radiological Protection; reports available through Pergamon Press, New York
NCRP	National Council on Radiation Protection and Measurement, Bethesda, MD
NRC	United States Nuclear Regulatory Commission, Washington D.C.

SUMMARY

- Stochastic effects are effects for which the probability of occurrence (and not the severity) is proportional to dose. The three stochastic effects that have been identified with radiation are:
 - Carcinogenesis
 - Teratogenesis
 - Mutagenesis
- The subcategories of acute radiation syndrome (ARS) are:
 - Hematopoietic
 - Gastrointestinal
 - Cerebrovascular
- Collective dose equivalent (CDE), the sum of the dose equivalents experienced by all the individuals in a population, has units of person-sievert (person-rem).
- Effective dose uses weighting factors to convert the equivalent dose delivered to specific body regions into a measure of overall harm to the organism and to future generations through genetic defects.
- Sources of natural background radiation, from largest to smallest are:
 - Radon
 - Internal
 - Terrestrial
 - Cosmic
- The yearly total equivalent dose from natural background is 3.0 mSv (300 mrem).
- Sources of man-made background radiation, from largest to smallest, are:
 - Medical x rays
 - Nuclear medicine
 - Consumer products
 - Other sources (occupational, nuclear power, industrial, fallout from nuclear weapons exploded in the atmosphere since 1945)
- The yearly total equivalent dose from man-made background is 0.6 mSv.
- Populations that have been exposed to unusual levels of radiation include:
 - Atom bomb survivors
 - Early radiologists, nurses, technologists (prior to World War II)
 - Uranium and other miners
 - Radium dial painters
 - Radiation therapy patients
 - Early diagnostic radiology patients
 - Populations exposed to fallout from nuclear weapons tests
 - Residents of regions with high natural background
 - Individuals participating in air and space travel
 - Victims of nuclear accidents
- A reasonable and widely accepted mathematical form that fits currently epidemiological data is the linear, nonthreshold model for most forms of cancer and the linear-quadratic, nonthreshold model for leukemia.
- A widely accepted “rule of thumb” for estimating the risk of fatal cancer per unit effective dose due to a single exposure to ionizing radiation is “5% per sievert.”

REFERENCES

1. Statkiewicz-Sherer, M. A., Visconti, P. J., and Ritenour, E. R. *Radiation Protection in Medical Radiography*, 4th edition. Chicago, Mosby, 2001.
2. International Commission on Radiological Protection (ICRP): *Nonstochastic Effects of Ionizing Radiation*, ICRP Publication 41. Oxford, Pergamon Press, 1984.
3. United Nations Scientific Committee on the Effects of Atomic Radiation (UNSCEAR): *Ionizing Radiation: Sources and Biological Effects*, Report E.82.IX.8. New York, United Nations, 1982.
4. “U.S. Food and Drug Administration: FDA Advisory: Avoidance of Serious X-Ray Induced Skin Injuries During Fluoroscopically-Guided Procedures,” Center for Devices and Radiological Health, U.S. Food and drug Administration, 8 September 1994.

5. O'Dea, J. O., Geise, R. A., and Ritenour, E. R. The potential for radiation-induced skin damage in interventional neuroradiological procedures: A review of 522 cases using automated dosimetry. *Med. Phys.* 1999; **26**(9):2027–2033.
6. Recommendations of the International Commission on Radiological Protection: *International Commission on Radiological Protection*, Publication 26. New York, Pergamon Press, 1977.
7. National Council on Radiation Protection and Measurements (NCRP) Report No. 93, Ionizing Radiation Exposure of the Population of the United States, Bethesda, MD, NCRP, 1987.
8. National Council on Radiation Protection and Measurements (NCRP) Report No. 94, Exposure of the Population in the United States and Canada from Natural Background Radiation, Bethesda, MD, NCRP, 1987.
9. National Council on Radiation Protection and Measurements (NCRP) Report No. 95, Radiation Exposure of the US Population from Consumer Products and Miscellaneous Sources, Bethesda, MD, NCRP, 1987.
10. United Nations Scientific Committee on the Effects of Atomic Radiation, Ionizing Radiation: Sources and Biological Effects, No.E.82.IX.8., 06300P (United Nations, New York), 1982.
11. Committee on the Biological Effects of Ionizing Radiations (BEIR V): *Health Effects of Exposure to low Levels of Ionizing Radiation*. National Research Council. Washington, D.C., National Academy Press, 1990, p. 162.
12. Shimizu, Y., Kato, H., Schull, W. J., et al. Life Span Study Report 11, Part 1. Comparison of Risk Coefficients for Site-Specific Cancer Mortality Based on the DS86 and T65DR Shielded Kerma and Organ Doses. Technical Report RERF TR 12-87. Hiroshima Radiation Effects Research Foundation, 1987.
13. Tokunaga, M., Land, C., Yamamoto, T., et al. Incidence of female breast cancer among atomic bomb survivors, Hiroshima and Nagasaki, 1950–1985. *Radiat. Res.* 1987; **112**:243–272.
14. Beebe, G., Kato, H., and Land, C. Mortality experience of atomic bomb survivors, 1950–1974 Life Span Study Report 8, Radiation Effects Research Foundation, 1978.
15. Shigematsu, I. The 2000 Sievert Lecture—Lessons from atomic bomb survivors in Hiroshima and Nagasaki. *Health Phys.* 2000; **79**(3):234–241.
16. Ishimaru, T., Cihak, R. W., Land, C. E., et al. Lung cancer at autopsy in A-bomb survivors and controls, Hiroshima and Nagasaki, 1961–1970 II. Smoking, occupation, and A-bomb exposure. *Cancer* 1975; **36**:1723–1728.
17. Jablon, S., and Kato, H. Childhood cancer in relation to prenatal exposure to atomic bomb radiation. *Lancet* 1970; **2**:1000–1003.
18. Seltser, R., and Sartwell, P. E. The influence of occupational exposure on the mortality of American radiologists and other medical specialists. *Am. J. Epidemiol.* 1965; **81**:2–22.
19. Matanoski, G., Seltser, R., Sartwell, P., et al. The current mortality rates of radiologists and other physician specialists. Deaths from all causes and from cancer. *Am. J. Epidemiol.* 1975; **101**:188–198.
20. Matanoski, G., et al. The current mortality rates of radiologists and other physician specialists. Specific causes of death. *Am. J. Epidemiol.* 1975; **101**:199–210.
21. Ulrich, H. The incidence of leukemia in radiologists. *N. Engl. J. Med.* 1946; **234**:45–46.
22. Lorenz, E. Radioactivity and lung cancer, a critical review of lung cancer in the miners of Schneeberg and Joachimsthal. *J. Natl. Cancer Inst.* 1944; **5**:1–13.
23. Whittemore, A. S., and McMillan, A. Lung cancer mortality among U.S. uranium miners. A reappraisal. *J. Natl. Cancer Inst.* 1983; **71**:489–499.
24. Archer, V. E., Wagoner, J. K., Lundin, F. E., Jr. Lung cancer among uranium miners in the United States. *Health Phys.* 1973; **25**:351–371.
25. Committee on the Biological Effects of Ionizing Radiations (BEIR IV). *Health Risks of Radon and Other Internally Deposited Alpha-Emitters*. National Research Council, Washington, D.C., National Academy Press, 1988, pp. 445–488.
26. Martland, H. S. The occurrence of malignancy in radioactive persons. *Am. J. Cancer*. 1931; **15**:2435–2516.
27. Spiers, F. W., Lucas, H. F., Rundo, J., et al. Leukemia incidence in the U.S. dial workers. *Health Phys.* 1983; **44**:65–72.
28. Stebbings, J. H., Lucas HF, Stehney AF: Mortality from cancers of major sites in female radium dial workers. *Am. J. Ind. Med.* 1984; **5**:435–459.
29. Rowland, R. E., Stehney, A. F., and Lucas, H. F., Jr. Dose-response relationships for female radium dial workers. *Radiat. Res.* 1978; **76**:368–383.
30. Polednak, A. P., Stehney, A. F., Rowland RE: Mortality among women first employed before 1930 in the U.S. radium dial-painting industry. *Am. J. Epidemiol.* 1978; **107**:179–195.
31. Darby, S. G., Doll, R., and Smith, P. G. Paper 9. Trends in long-term mortality in ankylosing spondylitis treated with a single course of x-rays. Health effects of low dose ionizing radiation. London, BNES, 1988.
32. Shore, R. E., Hildreth, N., Woodard, E., et al. Breast cancer among women given x-ray therapy for acute postpartum mastitis. *J. Natl. Cancer Inst.* 1986; **77**:689–696.
33. Court Brown, W. M., and Doll, R. Mortality from cancer and other causes after radiotherapy for ankylosing spondylitis. *Br. Med. J.* 1965; **2**:1327–1332.
34. Hempelmann, L., Hall, W., Phillips, M., et al. Neoplasms in persons treated with x-ray in infancy: Fourth survey in 20 years. *J. Natl. Cancer Inst.* 1975; **55**:519–530.
35. Modan, B., Ron, E., and Werner, A. Thyroid cancer following scalp irradiation. *Radiology* 1977; **123**:741–744.
36. Smith, P., and Doll, R. Late effects of x-irradiation in patients treated for metropathia haemorrhagica. *Br. J. Radiol.* 1976; **49**:224–232.
37. Boice, J. Jr., Rosenstein, M., and Trout, E. D. Estimation of breast doses and breast cancer risk associated with repeated fluoroscopic chest examinations of woman with tuberculosis. *Radiat. Res.* 1978; **73**:373–390.
38. Gibson, R., Graham, S., Lilienfeld, A., et al. Irradiation in the epidemiology of leukemia among adults. *J. Natl. Cancer Inst.* 1972; **48**:301–311.
39. Holm, L. E., Lundell, G., and Walinder, G. Incidence of malignant thyroid tumors in humans after exposure to diagnostic doses of iodine-131. Retrospective cohort study. *J. Natl. Cancer Inst.* 1980; **64**:1055–1059.
40. Harvey, G., Boice, J., Honeyman, M., et al. Prenatal x-ray exposure and childhood cancer in twins. *N. Engl. J. Med.* 1985; **312**:541–545.
41. Sherman, G. J., Howe, G. R., Miller, A. B., et al. Organ dose per unit exposure resulting from fluoroscopy for artificial pneumothorax. *Health Phys.* 1978; **35**:259–269.
42. Gunz, F., and Atkinson, H. Medical radiation and leukemia: A retrospective survey. *Br. Med. J.* 1964; **1**:389.
43. Gibson, R., Graham, S., Lilienfeld, A., et al. Irradiation in the epidemiology of leukemia among adults. *J. Natl. Cancer Inst.* 1972; **48**:301–311.
44. Committee on the Biological Effects of Ionizing Radiations (BEIR V). *Health Effects of Exposure to Low Levels of Ionizing Radiation*. National Research Council. Washington, D.C., National Academy Press, 1990, pp. 375–377.
45. Conard, R. A., et al. March 1958 Medical survey of Rongelap people. Brookhaven National Laboratory, Report 534 (T-135), 1959.
46. Conard, R. A., and Hicking, A. Medical findings in Marshallese people exposed to fallout radiation. *JAMA* 1965; **192**:457.
47. Barcinski, M. A., Abreu, M. D. C. A., De Almeida, J. C. C., et al. Cytogenetic investigation in a Brazilian population living in an area of high natural radioactivity. *Am. J. Hum. Genet.* 1975; **27**:802–806.
48. Kochupillan, N., Verma, I. C., Grewal, M. S., et al. Down's syndrome and related abnormalities in an area of high background radiation in coastal Kerala. *Nature* 1976; **262**:60–61.
49. Tao, Z., and Wei, L. An epidemiological investigation of mutational diseases in the high background radiation area of Yangjiang, China. *J. Radiat. Res.* 1986; **27**:141–150.
50. Jablon, S., and Bailar, S. C., III: The contribution of ionizing radiation to cancer mortality in the United States. *Prevent. Med.* 1980; **9**:219–226.
51. Pochin, E. E. Problems involved in detecting increased malignancy rates in the areas of high natural radiation background. *Health Phys* 1976; **31**:148–154.
52. Frigerio, N. A., and Stowe, R. S. Carcinogenic and genetic hazard from background radiation, in *Biological and Environmental Effects of Low-Level Radiation*. Vienna, International Atomic Energy Agency, 1976, pp. 385–393.
53. Vargo, G. J. A brief history of nuclear criticality accidents in Russia—1953–1997. *Health Phys* .1999; **77**(5):505–511.
54. Hess, D. B., and Hendee, W. R. Radiation and Political Fallout of Three Mile Island, in Hendee, W. R. (ed.), *Health Effects of Low-Level Radiation*. East Norwalk, CT, Appleton-Century-Crofts, 1984, pp. 247–258.
55. Normal, C., and Dickson, D. The aftermath of Chernobyl. *Science* 1986; **233**:1141–1143.
56. Committee on the Biological Effects of Ionizing Radiations (BEIR V): *Health Effects of Exposure to Low Levels of Ionizing Radiation*. National Research Council. Washington, D.C., National Academy Press, 1990, p. 23.
57. National Council on Radiation Protection and Measurements. *Influence of Doses and Its Distribution in Time on Dose-Response Relationships for Low LET*

- Radiations, Report No. 64. Bethesda, MD, National Council on Radiation Protection and Measurements, 1980.
58. Committee on the Biological Effects of Ionizing Radiations (BEIR V). *Health Effects of Exposure to Low Levels of Ionizing Radiation*. National Research Council. Washington, D.C., National Academy Press, 1990, p. 267.
 59. United Nations Scientific Committee on the Effects of Atomic Radiation (UNSCEAR). *Sources and Effects of Ionizing Radiation*, Report E. 77 IX. New York, United Nations, 1977.
 60. Committee on the Biological Effects of Ionizing Radiations (BEIR V). *Health Effects of Exposure to Low Levels of Ionizing Radiation*. National Research Council. Washington, D.C., National Academy Press, 1990, pp. 1–8.
 61. Committee on the Biological Effects of Ionizing Radiations (BEIR V). *Health Effects of Exposure to Low Levels of Ionizing Radiation*. National Research Council. Washington, D.C., National Academy Press, 1990, p. 165.
 62. Radiation Effects Research Foundation. *U.S.–Japan Joint Reassessment of Atomic Bomb Radiation Dosimetry in Hiroshima and Nagasaki—Final Report*, Vol. 1. Hiroshima, Japan, 1987.
 63. Radiation Effects Research Foundation: *U.S.–Japan Joint Reassessment of Atomic Bomb Radiation Dosimetry in Hiroshima and Nagasaki—Final Report*, Vol. 2. Hiroshima, Japan, 1988.
 64. Hendee, W. R. Estimation of radiation risks: BEIR V and its significance for medicine. *JAMA* **268**(5):620–624, 1992.

PROTECTION FROM EXTERNAL SOURCES OF RADIATION

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OBJECTIVES

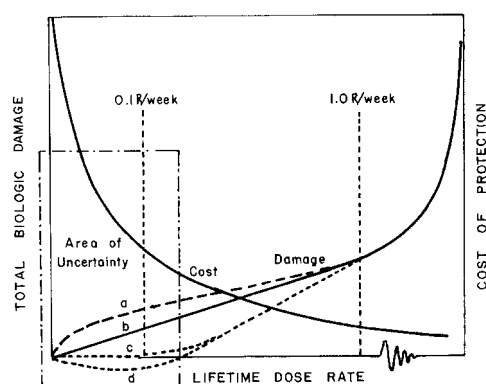
After completing this chapter, the reader should be able to:

- List the effective dose limits recommended by the NCRP.
- Discuss special regulations that concern pregnant workers.
- Convert monitor badge readings to effective dose.
- List some regulatory and some advisory groups in the field of radiation protection.
- Give an estimate for the radiation dose associated with radiologic procedures.
- Describe the methods used to calculate the amount of shielding required for various types of radiologic equipment.
- Explain the difference between
 - Primary and secondary barriers
 - Scatter and leakage exposure
 - Controlled and uncontrolled areas
- Write the fundamental equation used to calculate barrier thickness and explain each of its terms.
- Define the terms
 - Population dose
 - Genetically significant dose
 - Gonadal dose

For a few years after their discovery, x rays and radioactive materials were used with little regard for their biological effects. After a few years, the consequences of careless handling and indiscriminate use of radiation sources became apparent. These consequences included severe burns, epilation, and, later, leukemia and other forms of cancer in persons who received high exposures to radiation. Persons affected included many who pioneered the medical applications of ionizing radiation.

Because ionizing radiation had proved beneficial to humans in many ways, the question to be answered was whether individuals in particular, and the human race in general, could enjoy the benefits of ionizing radiation without suffering unacceptable consequences in current or future generations. This question is often described as the problem of *risk versus benefit*. To reduce the risk of using ionizing radiation, advisory groups were formed to establish upper limits for the exposure of individuals to radiation. Advisory groups have reduced the upper limits recommended for radiation exposure many times since the first limits were promulgated. These reductions reflect the use of ionizing radiation by a greater number of persons, the implications of new data concerning the sensitivity of biological tissue to radiation, and improvements in the design of radiation devices and in the architecture of facilities where persons use radiation devices.

The philosophy underlying the control of radiation hazards is a *philosophy of risk*. With this philosophy, advisory groups attempt to establish upper limits for radiation exposure that minimize the hazard to individuals and to the population but do not interfere greatly with the beneficial uses of radiation.¹ This philosophy of risk is depicted in Margin Figure 28-1. The total biological damage to a population is expressed as the sum of individual effects, such as reduced vitality, morbidity, shortened life span, and genetic damage, that may result from receipt of some average dose rate over the lifetime of each individual in the population. The total biological damage is assumed to increase gradually as the average dose rate increases to a value of perhaps 0.01 Sv (1 rem) per week. Damage is assumed to increase more rapidly beyond this dose rate. Although the exact shape is unknown for the curve of total biological damage versus dose rate, the curve in Margin Figure 28-1 is probably a reasonable estimate. The region of the curve enclosed within the rectangle labeled the “Area of Uncertainty” is the region of greatest interest to persons concerned with radiation protection; however, this part of the curve is also the region for which few data are available. As indicated by curve c, the damage may remain at zero for average dose



MARGIN FIGURE 28-1

Total biologic damage to a population expressed as a function of the average dose rate to individuals in the population. The cost of protection is reduced as greater biologic damage is tolerated. (From Claus, W. The Concept and Philosophy of Permissible Dose to Radiation, in Claus, W. (ed.), *Radiation Biology and Medicine*. Reading, MA, Addison-Wesley, 1958. Used with permission.)

rates below some threshold value. Conceivably, the curve for total biological damage may fall below the axis near the origin (curve d), which suggests that very low dose rates are beneficial, a concept known as radiation hormesis. Limited experimental data support this hypothesis. Perhaps the curve rises rapidly at the origin, as suggested in curve a. The data are meager for the biological consequences of radiation exposure at low dose rates, and curve b is usually assumed to be correct. This curve suggests that the total biological damage to a population is linearly proportional to the average dose rate down to a dose rate of zero for individuals in the population. This suggestion is referred to as the *linear nonthreshold hypothesis of radiation injury*.

Adequate data are not available for the area of uncertainty depicted in Margin Figure 28-1. Consequently, the cost of radiation protection must be balanced against unknown biological effects that might result if adequate protection were not provided. The cost of protecting individuals from ionizing radiation (e.g., by shielding, remote control techniques, monitoring procedures, and personnel restriction) is plotted as a function of the average dose rate acceptable to the population. The cost increases from almost zero with no restrictions on radiation exposure, to a cost that approaches infinity if the population desires no exposure to radiation. Somewhere within the area of uncertainty, an upper limit must be accepted for the exposure of persons to radiation. This limit should be recognized as an upper limit for radiation exposure that provides a risk acceptable to the population, without depriving the population of benefits derived from the judicious use of ionizing radiation. In practice, the exposure of individuals to radiation should be kept as low as reasonably achievable.

■ REGULATORY AUTHORITY FOR RADIATION PROTECTION

Various agencies play some role in the limitation of radiation dose to the general public and to occupationally exposed individuals. There are two types of agencies, advisory and regulatory. Advisory agencies assemble teams of experts in epidemiology, public health, radiobiology, and other fields to publish analyses of the world's literature on radiation bioeffects and to recommend guidelines, limits, and procedures that would lead to the most effective use of radiation for the public good while minimizing the hazard. Examples of advisory agencies include:

- National Council on Radiation Protection and Measurements (NCRP)
- International Commission on Radiological Protection (ICRP)
- United Nations Scientific Committee on the Effects of Atomic Radiation (UNSCEAR)
- National Academy of Sciences (NAS)
- Conference of Radiation Control Program Directors (CRCPD)

Regulatory agencies make laws, inspect facilities, review records, levy fines, and issue and rescind licenses. The rules that they promulgate generally follow the guidelines of the advisory groups, usually with some time delay. Examples of regulatory agencies (and their jurisdictions) include:

- Nuclear Regulatory Commission (NRC)—radioactive materials used or produced in nuclear reactors (source and by product radioactive material).
- State Health Departments—x-ray producing equipment and, in some states, radioactive materials.
- Environmental Protection Agency (EPA)—radioactive materials and sources that affect the environment.
- Food and Drug Administration (FDA)—manufacture of X-ray producing equipment.

State Health Departments in some states have established their own regulations for radiation protection that are equivalent to or stricter than those of the NRC. Through

The responsibility for recommending limits for radiation exposure has been delegated to advisory groups composed of persons experienced in the use of ionizing radiation. In 1922, the American Roentgen Ray Society and the American Radium Society began a study of acceptable levels of radiation exposure. In 1928, the International Congress of Radiology formed the International Commission on Radiological Protection, referred to commonly as the ICRP. This organization is recognized as the international authority for the safe use of sources of ionizing radiation. The U.S. Advisory Committee on X-Ray and Radium Protection, known now as the National Council on Radiation Protection and Measurements (the NCRP), was formed in 1929 to interpret and implement recommendations of the ICRP for the United States.

TABLE 28-1 Effective Dose Limits Recommended by the NCRP

Occupational exposures	
Effective dose limits	
(a) Annual	50 mSv
(b) Cumulative	10 mSv × age
Equivalent dose limits for tissues and organs (annual)	
(a) Lens of eye	150 mSv
(b) Skin, hands, and feet	500 mSv
Guidance for emergency occupational exposure	500 mSv
Public exposures (annual)	
Effective dose limit, continuous or frequent exposure	1 mSv
Effective dose limit, infrequent exposure	5 mSv
Equivalent dose limits for tissues and organs	
(a) Lens of eye	15 mSv
(b) Skin, hands, and feet	50 mSv
Education and training exposures (annual)	
Effective dose limit	1 mSv
Equivalent dose limit for tissues and organs	
(a) Lens of eye	15 mSv
(b) Skin, hands, and feet	50 mSv
Embryo–fetus exposures (monthly)	
Equivalent dose limit	0.5 mSv

an agreement with the NRC, these states, known as agreement states, have accepted authority to regulate radioactive byproduct material within their borders.

There is no one set of regulations for radiation protection. The regulations that apply in a particular facility depend upon what state it is in, whether it is a federal facility, whether it produces emissions that fall under the purview of the EPA, and so on. Many of the agreement states do have similar regulations, however. The Conference of Radiation Control Program Directors is an organization that is composed of State Health Department representatives. They produce a document, known as the Suggested State Regulations for the Control of Radiation, that is often adopted with minor changes by individual states. Also, the U.S. Code of Federal Regulations, although technically binding only upon federal facilities and non-agreement states, is often adopted in whole or in part by individual states.

To reduce confusion and redundancy, the dose limits and radiation safety procedures presented in this chapter are those that are recommended by the NCRP and the ICRP. Those recommendations are at any time similar to or slightly ahead of implementation schedules of regulatory authorities and therefore represent the most up-to-date information.

EFFECTIVE DOSE LIMITS

For the purpose of developing recommendations for the upper limits of exposure of personnel to ionizing radiation, the ICRP and NCRP assume that a population is divided into two main groups: occupationally exposed individuals and members of the general public.^{2,3}

Radiation Dose to Occupationally-Exposed Persons

For persons whose occupation requires exposure to radiation, recommendations concerning limits to the effective dose* (E) are provided in Table 28-1. In the past, the expression maximum permissible dose (MPD) was used to describe regulations regarding personnel exposure. This expression has now been replaced by the term “effective dose limit.” Effective dose limits do not include exposures from natural

*Effective dose is defined in Chapter 6.

background radiation or medical exposures when the worker is a patient. Medical exposures involve risk/benefit decisions regarding the patient, and are fundamentally the prerogative of the patient and or family, with guidance from the physician. However, guidelines for such decisions have been promulgated by the U.S. Food and Drug Administration,⁴ the U.S. Nuclear Regulatory Commission, and the Environmental Protection Agency.⁵

The effective dose limit for occupationally exposed individuals is 50 mSv (5 rem) per year. As defined in Chapter 6, calculation of effective dose involves the use of both tissue-weighting factors and radiation weighting factors to arrive at a measure of overall somatic and genetic detriment. Because some organs have a low tissue-weighting factor, it is possible that deterministic effects could occur in such an organ while the effective dose to the individual is below the limit. Therefore, separate limits are set for the lens of the eye (150 mSv) and localized areas of the skin, hands, and feet (500 mSv). These limits are expressed as equivalent doses rather than effective doses because the intention is not to determine the overall somatic and genetic detriment caused by the organ dose, but to specify the organ dose itself. The lens of the eye, skin, hands, and feet are the only organs for which the special deterministic dose limits apply. All other organs are considered by the NCRP to be adequately protected by the effective dose limit.⁶

Most occupationally exposed persons are able to keep their exposures far below the effective dose limit. To ensure that the lifetime risk of these individuals is acceptable, the NCRP recommends that the cumulative lifetime occupational effective dose in millisieverts should not exceed ten times the age of the worker in years. For example, a 50-year-old worker exposed to the yearly effective dose limit each year since beginning work at the age of 18 years would have received a total of $50(50 - 18) = 1600$ mSv (160 rem). This exposure violates the cumulative lifetime limit of 10 times the age, that is, 500 mSv (50 rem). The purpose of the cumulative limit is to control the lifetime total effective dose as well as the yearly effective dose, and to encourage ongoing evaluation of this number to keep career exposures to a “reasonable” level. In short, the intent is to keep actual yearly effective doses well below the yearly limits.

A number of special cases arise when exposure limits are applied to particular individuals. These special cases include the following:

1. **The Pregnant Worker and Protection of the Embryo–Fetus.** An equivalent dose limit of 0.5 mSv per month is recommended for the fetus once the pregnancy is known. This amount does not include medical exposure or natural background radiation.

When a pregnant worker notifies her employer of her pregnancy, the radiation safety officer should review her badge readings over recent months to estimate what the fetal dose would have been during that time period. Because of the margin of safety usually employed in designing the workplace to limit the worker’s actual exposure, there is seldom any need to reassign a worker or make any change in her activities. To reassure all parties during a pregnancy, some institutions routinely issue extra monitoring devices that are read more frequently or have an audible alarm if a preset dose threshold is reached.

Because of the legal outcome of “right to work” lawsuits brought on behalf of female workers in industries where hazardous materials may be encountered (e.g., lead and other heavy metals), most radiation safety regulations require a woman to notify her employer that she is pregnant before any special actions are taken. The institution does not have the authority to make changes in activities, responsibilities, record keeping, monitoring, or counseling unless or until the employee notifies the employer that she is pregnant.

There are no separate specific controls for occupationally-exposed women who are not known to be pregnant. Although reports of the United Nations Scientific Committee on the Effects of Atomic Radiation⁷ estimate that the lifetime cancer risk from radiation received during the fetal stage is two to three times that of the adult (i.e., approximately 10% per Sv), the effective dose experienced by the fetus in a

In 1991 the ICRP revised their recommendations on radiation protection to incorporate recent information from the U.S. National Academy of Sciences and from the United Nations Scientific Committee on the Effects of Atomic Radiation. This revision involved a reanalysis of the risk of cancer and genetic effects. The reanalysis was possible because of declassification of atomic bomb yield data and because of the appearance of cancers in the atomic bomb survivor population that occurred later in the life of the individuals who were exposed to unusual amounts of radiation 46 years earlier. The main change was the reformulation of tissue weighting factors (see Chapter 6). Their recommendations are considered more inclusive not only because of the new data, but also because they took into account the risk of severe hereditary effects in all future generations (previous formulations had concerned only the first two generations) and the risk of nonfatal cancers. However, the most recent federal regulations adopted by the Nuclear Regulatory Commission in 1991 had already been through lengthy revision cycles, including public comment periods. These regulations were adopted using older terminology and weighting factors. At the present time, when a radiation safety officer of a facility calculates dose from personal dosimeter records, the results are expressed in terms of the older “effective dose equivalent” system which uses earlier versions of weighting factors and an equation that is slightly different from Eq. (28-9) and (28-10). Because of the large margins of safety that are built into regulations, the discrepancy has little effect on actual safety concerns. However, it is expected that there will be revisions in the future that bring Federal and State regulations into alignment with recommendations of the scientific advisory groups.

controlled occupational environment would be small. In the unlikely scenario of a fetus receiving as much as 50 mSv prior to discovery of a pregnancy, the lifetime cancer risk to the offspring would be estimated to be at most $(0.05 \text{ Sv}) (10\% \text{ Sv}^{-1}) = 0.5\%$ lifetime risk.

2. Education and Training Exposures. Exposure of individuals, even if under the age of 18, is permitted for purposes of education and training. This includes demonstrations for K-12 classrooms as well as for radiologic technology students who have not been formally employed.

It is recommended that education and training exposures be limited to an effective dose of less than 1 mSv per year. This limit is intended to count as part of the 5 mSv per year limit for “infrequent exposure” of members of the general public discussed in a later section. The equivalent dose to the lens of the eye is limited to 15 mSv annually, and the equivalent dose to the skin, hands, and feet is limited to 50 mSv annually.

3. Emergency Occupational Exposure. Many emergency situations may be handled within occupational limits for radiation exposure. In cases of extreme emergencies such as lifesaving events, however, there may be occasions when the risk versus benefit analysis favors a one-time overexposure of individuals to achieve a greater good. In such situations, most advisory groups recommend using older volunteers because most stochastic effects of radiation exposure are more likely to occur if the exposure is received at an earlier age.

In an emergency situation, the National Council on Radiation Protection and Measurements recommends that measures be taken to ensure that the emergency exposure is less than 500 mSv. It is further recommended that the equivalent dose to the skin be kept below 5 Sv. The latter requirement applies to exposures from low-energy photons or beta particles during lifesaving actions. In such circumstances, the tolerance dose for skin damage could be reached before the 500 mSv effective dose for the whole body was exceeded.

If exposures exceed the limits for an individual by a small amount, the expectation of harm to that individual is small. Nevertheless, this occurrence should be investigated so that the reason for exceeding the limit may be ascertained. Questions such as whether there was a lapse in institutional safety procedures, and what will be done to prevent this occurrence in the future, must be answered. To facilitate investigation of problem areas before significant exposures occur, institutions set their own action levels. Action levels are levels that are well below effective dose limits but higher than exposures necessary to perform routine occupational tasks. Many institutions set action levels at one-tenth of the effective dose limits, and they monitor the equivalent monthly or quarterly readings of personnel exposures. If the action level is exceeded but the effective dose limit has not been reached, an internal investigation is conducted without necessarily altering the individual's work conditions.

The guiding philosophy of radiation protection is that it is not desirable for personnel to routinely reach an effective dose limit. When designing facilities and in developing policies and procedures, the goal is to ensure that exposures of personnel are maintained as low as reasonably achievable (ALARA). While the definition of what is “reasonable” may vary from one situation to another, the principle of ALARA mandates consideration of measures beyond those strictly required to maintain effective dose limits. For example, the cost of adding extra shielding in the walls of examination rooms, or the modification of policies to provide further restrictions on exposure time or increased distance from radioactive sources, may be part of an ALARA program.^{2,8}

Radiation Dose to Members of the Public

Personal monitors are not feasible for members of the public exposed occasionally to radiation. The effectiveness of protective measures for these persons is evaluated indirectly by sampling the air, water, soil, and other elements of the environment. The expected dose for members of the public is computed from analyses of the

environment and from knowledge of the living habits of the members. The effective dose limit for members of the public excludes the radiation dose contributed by background radiation and by medical exposures, and in general it is one-tenth of the whole-body effective dose limit for occupationally exposed persons. That is, the effective dose limit for a whole-body dose is 5 mSv/yr (0.5 rem/yr) for occasional exposure of members of the public. If the exposure is frequent or continuous, the limit is lowered to 1 mSv/yr (0.1 rem/yr).

Radiation exposure limits for the general public do not include medical exposures. That is, the 10 to 50 mSv (1 to 5 rem) received from a computed tomographic (CT) scan does not “exceed” the dose limit for the patient. Such an exposure has nothing to do with the limits discussed here. Medical procedures are performed on the basis of a risk/benefit analysis for the individual and are considered to be separate from the coincidental radiation exposures discussed in this section.

Population Exposure, Gonadal Dose, and Genetically Significant Dose

In addition to recommending limits for the exposure of individuals, scientific advisory groups periodically estimate the actual impact of radiation exposure upon a population of persons. For this purpose, the *collective effective dose* is defined as the average dose estimated for individuals in the population multiplied by the total number of individuals in the population. The current estimate of collective effective dose for the U.S. population is 835,000 person-Sv. This estimate is the product of the average annual background of 3.6 mSv times the U.S. population, which, at the time of the most recent estimate, was 230,000,000.

The exposure of gonadal tissue in a large population is of concern because of the possibility of disrupting a significant fraction of the gene pool. The approximate dose to gonadal tissue that is delivered by various medical procedures is given in Table 28-2. In a medical facility, the output of each radiation-producing device should be measured yearly, and these measurements should be used to compute patient doses, including gonadal doses for each medical procedure in which the device is used. A prediction of the “genetic impact” of such exposures is based upon estimates of the

TABLE 28-2 Typical Gonad Doses from Various Radiographic Examinations

X-Ray Examination	Gonad Dose (mrad) ^a	
	Male	Female
Skull	<1	<1
Cervical spine	<1	<1
Full-mouth dental	<1	<1
Chest	<1	<1
Stomach and upper gastrointestinal	2	40
Gallbladder	1	20
Lumbar spine	175	400
Intravenous urography	150	300
Abdomen	100	200
Pelvis	300	150
Upper limb	<1	<1
Lower limb	<1	<1

^aFor some radiologic examinations the female gonad dose is greater than the dose received by the male because the female reproductive organs are located within the pelvic cavity, unlike the male reproductive organs, which are located outside and below the pelvic cavity. The distribution of biologic tissue overlying the ovaries also affects the dose received for a given radiologic examination.

Source: From Ballinger, P. W. *Merrill's Atlas of Radiographic Positions and Radiologic Procedures*, Vol. 1, 8th edition. St. Louis, Mosby, 1995.

fraction of the population that undergoes diagnostic and nuclear medicine procedures each year, in addition to all other sources of exposure. Furthermore, the impact is weighted by the age and sex distribution of the population to take into account the expected number of future children that would be produced by each category. The *genetically significant dose* (GSD) is a quantity that is calculated in this manner. It is the dose that, if received equally by all members of the population, would result in the same genetic impact as the uneven distribution of gonadal exposures that actually occurs. Current estimates reveal a GSD of the U.S. population of 1.3 mSv annually. Diagnostic x-ray exposure accounts for 0.2 mSv and nuclear medicine accounts for 0.02 mSv of the total. The rest is attributable to background radiation exposure.

■ SAFETY RECOMMENDATIONS FOR SOURCES OF X AND γ RADIATION

Using recommendations of the ICRP for guidance, members of the NCRP have suggested a number of guidelines for evaluating the safety of x-ray equipment and sources of γ radiation. A complete description of these recommendations is available in a report of the NCRP.⁹

■ PROTECTIVE BARRIERS FOR RADIATION SOURCES

All occupations are associated with some risk of morbidity and mortality. Occupations in trade, service, and government are considered, in an actuarial sense, to be “safe” and are associated with risks of fatality of the order of 1 in 10,000 per year. Occupations such as construction, mining, and agriculture are associated with risks that are 3–5 times higher and are considered to be “more hazardous,” although still within a safe range. Because of the assumption that the risk of stochastic effects following exposure to ionizing radiation is nonthreshold, radiation safety regulations cannot be set so low that there is no theoretical risk. The National Council on Radiation Protection and Measurements uses a process of recommending limits based upon the goal of maintaining the risk of genetic effects, as well as fatal and nonfatal cancers, at the same level as that of “safe” industries. Estimates of the NCRP show that the lifetime risk from an average worker’s exposures over a working career is approximately 1 in 10,000. A worker who receives the maximum cumulative limit of 10 mSv/yr over an entire career would have a risk of approximately 5 in 10,000.

Occupational exposure should be monitored for each individual radiation worker. Any area where a yearly whole-body effective dose of 15 mSv (1.5 rem) or more could conceivably be delivered to an individual should be considered a controlled area and supervised by a radiation protection officer. Persons working in a controlled area and who are exposed to radiation should carry one or more personal monitors for the measurement of radiation dose (e.g., a film badge, thermoluminescent dosimeter, or pocket ionization chamber), and access to the area should be restricted.

The walls, ceiling, and floor of a room containing an x-ray machine or radioactive source may be constructed to permit the use of adjacent rooms when the x-ray machine is energized or the radioactive source is exposed. With an effective dose limit of 50 mSv/yr (5 rem/yr) for occupational exposure, the maximum permissible average dose equivalent per week equals 1.0 mSv (0.1 rem). Therefore, a dose of 1.0 mSv/wk (0.1 rem/wk) (or less) is used when computing the thickness of radiation barriers required for controlled areas. For areas outside the supervision of a radiation safety officer, a value of 0.1 mSv (0.01 rem) per week is usually used. Because the conversion from roentgens to rads is nearly 1 for x and γ rays (see Chapter 6) and because the quality factor is also 1 for these radiations, the maximum permissible exposure is often taken as 2.5×10^{-5} C/(kg-wk) (0.1 R/wk) or 2.5×10^{-6} C/(kg-wk) (0.01 R/wk) for computation of the barrier thicknesses for sources of x and γ rays. In a shielded room, *primary barriers* are designed to attenuate the primary (useful) beam from the radiation source, and *secondary barriers* are designed to reduce scattered and leakage radiation from the source.

Protection from Small Sealed γ -Ray Sources

The exposure X in roentgens* to an individual in the vicinity of a point source of radioactivity is

$$X = \frac{\Gamma_{\infty} A t}{d^2} (B) \quad (28-1)$$

where X is the exposure in roentgens, Γ_{∞} is the exposure rate constant in (R-m²)/(hr-Ci) at 1 m, A is the activity of the source in curies, t is the time in hours spent

*Shielding calculations are usually performed by using traditional units (e.g., roentgen, rad, rem) at the present time.

in the vicinity of the source, d is the distance from the source in meters, and B is the fraction of radiation transmitted by a protective barrier between the source and the individual. The exposure may be reduced by (1) increasing the distance d between the source and the individual, (2) decreasing the time t spent in the vicinity of the source, and (3) reducing the transmission B of the protective barrier (Margin Figure 28-2).

Example 28-1

What is the thickness of lead that reduces the exposure to 0.1 R/wk for a person at a distance of 1 m from a 100 mg radium source encapsulated with 0.5 mm Pt(Ir) for 40 hr/wk? The exposure rate constant Γ_{∞} is 0.825(R-m²)/(hr-Ci) for radium filtered by 0.5 mm Pt(Ir).

The 100-mg radium source is 0.1 g, which for radium is approximately 0.1 Ci.

$$\begin{aligned} X &= \frac{\Gamma_{\infty} A t}{d^2} (B) \\ B &= \frac{X d^2}{\Gamma_{\infty} A t} \\ &= \frac{(0.1)(1)^2}{(0.825)(0.1)(40)} \\ &= 0.03 \end{aligned} \quad (28-1)$$

From Margin Figure 28-2, about 7 cm of lead is required for the radiation barrier to yield a transmission of 0.03 (3%).

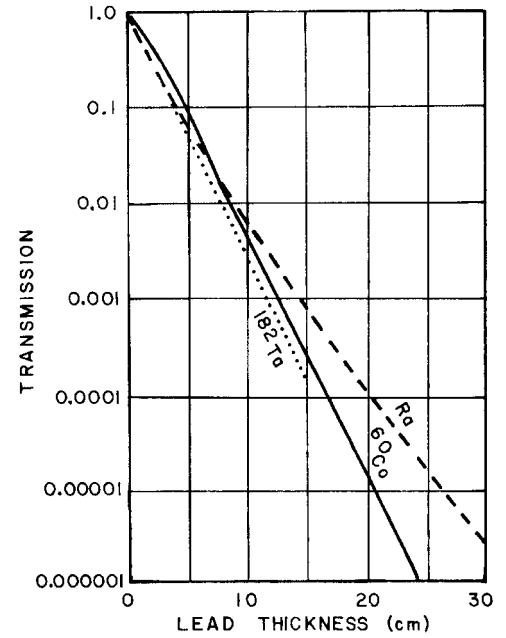
Primary Barriers for X-Ray Beams

For an individual some distance from an x-ray unit, the maximum expected exposure to radiation depends on the following:

1. Emission rate of radiation from the unit. As a general rule, a diagnostic x-ray unit operating at 100 kVp furnishes an exposure rate of less than 2.58×10^{-4} C/(kg-min) at 1 m for each milliamperere of current flowing through the x-ray tube. That is, the exposure rate is $\leq 2.58 \times 10^{-4}$ C/(kg-mA-min) or ≤ 1 R/(mA-min) at 1 m.
2. Average tube current and operating time of the x-ray unit per week. The product of these variables has units of milliamperere-minutes per week and is termed the *work load* W of the x-ray unit. Typical work loads for busy radiographic installations are described in Table 28-3.
3. Fraction of operating time that the x-ray beam is directed toward the location of interest. This fraction is termed the *use factor* U . Typical use factors are listed in Table 28-4.
4. Fraction of operating time that the location of interest is occupied by the individual. This fraction is termed the *occupancy factor* T . Typical occupancy factors are listed in Table 28-5. For controlled areas, the occupancy factor always is 1.
5. Distance d in meters from the x-ray unit to the location of interest because the exposure decreases with the inverse square of the distance ($1/d^2$).

The maximum expected exposure to an individual at the location of interest may be estimated as

$$\begin{aligned} \text{Maximum expected exposure} &= \frac{(1\text{R-m}^2/\text{mA-min})(\text{mA-min/wk}) \left(\frac{\text{Use}}{\text{factor}} \right) \left(\frac{\text{Occupancy}}{\text{factor}} \right)}{d^2} \quad (28-2) \\ &= \frac{WUT}{d^2} \end{aligned}$$



MARGIN FIGURE 28-2

Transmission through lead of γ rays from radium, ^{60}Co , and ^{226}Ra . (From International Commission on Radiological Protection. *Report of Committee III on Protection Against X Rays Up to Energies of 3 MeV and Beta and Gamma Rays from Sealed Sources*, ICRP Publication No. 3. New York, Pergamon Press, 1960. Used with permission.)

Radiation protection may be reduced to three fundamental principles: time, distance, and shielding. Radiation exposure may be reduced by decreasing the exposure time, increasing distance from the source of radiation, or providing more shielding between the source and exposed individuals.

Occupancy factors are always equal to one for controlled areas because it is assumed that if a given worker is not in one controlled area during a workday, he or she is likely to be in another controlled area.

TABLE 28-3 Typical Weekly Work Loads for Busy Installations

Diagnostic	Daily Patient Load	Weekly Work Load (W), mA-min ^a		
		100 kV or less	125 kV	150 kV
Chest (14 × 17:3 films per patient, no grid)	60	150	—	—
Cystoscopy	8	600	—	—
Fluoroscopy including spot filming	24	1500	600	300
Fluoroscopy without spot filming	24	1000	400	200
Fluoroscopy with image intensification including spot filming	24	750	300	150
General radiography	24	1000	400	200
Special procedures	8	700	280	140

^a Peak pulsating x-ray tube potential.

Source: National Council on Radiation Protection and Measurements. *Structural Shielding Design and Evaluation for Medical Use of X Rays and Gamma Rays of Energies Up to 10 MeV*. Recommendations of the NCRP, Report No. 49. Washington, D.C., 1976.

This exposure can be reduced to a maximum permissible level X_p (e.g., 0.1 or 0.01 R/wk) by introduction of an attenuation factor B into Eq. (28-2).

$$X_p = \frac{WUT}{d^2}(B)$$

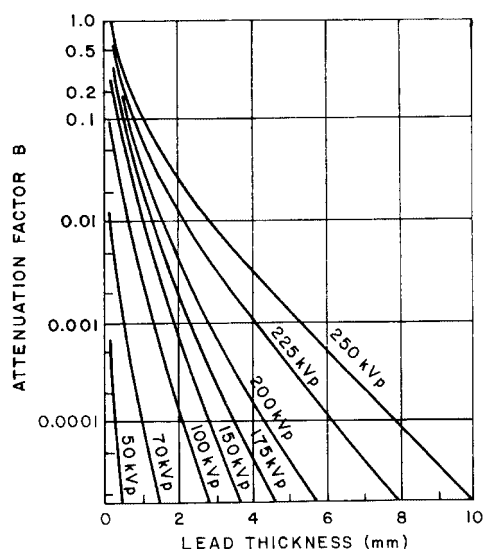
The attenuation factor B is

$$B = \frac{X_p d^2}{WUT} \quad (28-3)$$

With the attenuation factor B and the curves in Margin Figure 28-3, the thickness of concrete or lead may be determined that reduces the exposure rate to an acceptable level. Listed in Table 28-6 are the thicknesses of lead that are equivalent to concrete walls of different thicknesses.¹⁰ Half-value layers for other commonly used shielding materials are given in Table 28-7.

Example 28-2

A diagnostic x-ray generator has a busy work load of 1000 mA-min/wk at 100 kVp. The x-ray tube is positioned 4.5 m from a wall between the radiation room and a

**MARGIN FIGURE 28-3**

Attenuation in lead of x rays generated at tube voltages from 50 to 250 kVp. The curves were obtained with a half-wave-rectified x-ray generator and with a 90-degree angle between the electron beam and the axis of the x-ray beam. The inherent filtration was 3 mm of aluminum for the 150-kVp to 250-kVp curves and 0.5 mm of aluminum for the other curves. X-ray beams generated with a constant tube voltage require barriers 10% thicker than those indicated by the curves. (From International Commission on Radiological Protection. *Report of Committee III on Protection Against X Rays Up to Energies of 3 MeV and Beta and Gamma Rays from Sealed Sources*, ICRP Publication No. 3. New York, Pergamon Press, 1960. Used with permission.)

TABLE 28-4 Use Factors Recommended by the ICRP

Full use ($U = 1$)	Floors of radiation rooms except dental installations, doors, walls, and ceilings of radiation rooms exposed routinely to the primary beam.
Partial use ($U = \frac{1}{4}$)	Doors and walls of radiation rooms not exposed routinely to the primary beam; also, floors of dental installations.
Occasional use ($U = \frac{1}{16}$)	Ceilings of radiation rooms not exposed routinely to the primary beam. Because of the low use factor, shielding requirements for a ceiling are usually determined by secondary rather than primary beam considerations.

Source: International Commission on Radiological Protection. *Report of Committee III on Protection Against X Rays Up to Energies of 3 MeV and Beta and Gamma Rays from Sealed Sources*. ICRP Publication No. 3. New York, Pergamon Press, 1960. Used with permission.

TABLE 28-5 Occupancy Factors Recommended by the ICRP

Full occupancy ($T = 1$)	Control spaces, offices, corridors, and waiting spaces large enough to hold desks, darkrooms, workrooms and shops, nurse stations, rest and lounge rooms used routinely by occupationally exposed personnel, living quarters, children's play areas, and occupied space in adjoining buildings.
Partial use ($T = \frac{1}{4}$)	Corridors too narrow for desks, utility rooms, rest and lounge rooms not used routinely by occupationally exposed personnel, wards and patients' rooms, elevators with operators, and unattended parking lots.
Occasional occupancy ($T = \frac{1}{16}$)	Closets too small for future occupancy, toilets not used routinely by occupationally exposed personnel, stairways, automatic elevators, pavements, and streets.

Source: International Commission on Radiological Protection. *Report of Committee III on Protection Against X Rays Up to Energies of 3 MeV and Beta and Gamma Rays from Sealed Sources*, ICRP Publication No. 3. New York, Pergamon Press, 1960. Used with permission.

radiologist's office. The wall contains 3 cm of ordinary concrete. For a use factor of 0.5, what is the thickness of lead that must be added to the wall?

$$B = \frac{X_p d^2}{WUT} \quad (28-3)$$

Because the radiologist's office is a controlled area for occupationally exposed persons, $X_p = 0.1$ R/wk, $T = 1$, and

$$\begin{aligned}
 B &= \frac{0.1 d^2}{WUT} \\
 &= \frac{(0.1)(4.5)^2}{(1.000)(0.5)(1)} \\
 &= 0.004
 \end{aligned}$$

From the 100-kVp curve in Margin Figure 28-3, the thickness of lead required is 1.0 mm. A 3-cm thickness of concrete is approximately equivalent to 0.4 mm of lead

TABLE 28-6 Thicknesses of Lead Equivalent to Thicknesses of Ordinary Concrete (Density, 22 g/cm³) at Selected Tube Voltages for Broad X-Ray Beams

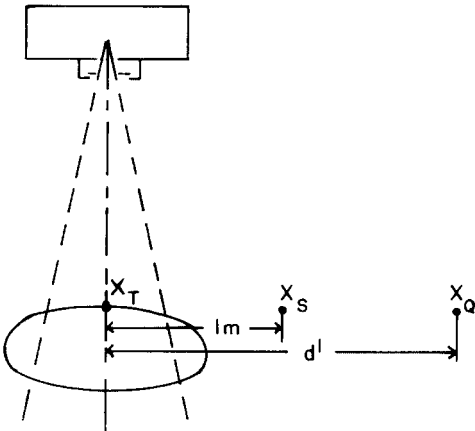
Thickness of Concrete (cm)	Lead Equivalent (mm) for X-Rays Generated at the Following Voltages (kVp)										
	50	75	100	150	200	250	300	400	500	1000	2000
5	0.4	0.5	0.6	0.5	0.5	0.6	0.8	1.1	1.6	4.0	6
10	0.9	1.2	1.4	1.2	1.2	1.7	2.2	3.0	3.9	8.6	13
15	1.4	2.0	2.4	1.9	2.1	3.0	3.8	5.4	7.1	13	22
20	2.0	2.8	3.4	2.7	2.9	4.4	5.8	8.5	11	21	31
25	2.5	3.6	4.4	3.4	3.8	5.8	7.9	11	15	29	40
30	3.1	4.3	5.4	4.2	4.7	7.3	10	14	19	37	49
35	—	—	—	5.1	5.6	8.6	12	18	24	45	58
40	—	—	—	—	—	—	—	21	28	54	67
45	—	—	—	—	—	—	—	24	33	62	76
50	—	—	—	—	—	—	—	—	37	71	85
60	—	—	—	—	—	—	—	—	46	88	103
75	—	—	—	—	—	—	—	—	60	112	130
90	—	—	—	—	—	—	—	—	—	138	159

Source: Appleton, G., and Krishnamoorthy, P. *Safe Handling of Radioisotopes; Health Physics Addendum*. Vienna, International Atomic Energy Agency, 1960. Used with permission.

TABLE 28-7 Half-Value Layers for Commonly Encountered Shielding Materials^a

Material	Average Half-Value Layers			
	50 kVp	70 kVp	100 kVp	125 kVp
Gypsum wallboard	14.12	28.03	HVLs (mm) 46.64	47.53
	±0.45	±0.30	±4.85	±1.80
Steel	0.27	0.74	1.46	2.11
	±0.03	±0.03	±0.08	±0.04
Plate glass	5.26	9.56	15.25	17.37
	±0.32	±0.33	±1.14	±1.22
Concrete			12.83	15.61
			±0.08	±0.06

^aData from Rossi, R. P., Ritenour, R., and Christodoulou, E. Broad beam transmission properties of some common shielding materials for use in diagnostic radiology. *Health Physics* 1991; **61**(5):601–608.



MARGIN FIGURE 28-4
Geometry for determining the radiation exposure X_Q at a distance d' from a scatterer where the exposure is X_T . The exposure X_S at a distance of 1 m from the scatterer is $1/1000 X_T$.

(Table 28-6). Consequently, at least 0.6 mm (or 1/32 in.) of lead should be added to the wall.

Secondary Barriers for Scattered X-Radiation

Radiation may be scattered to a location that is not exposed to the primary x-ray beam. The amount of scattered radiation varies with the intensity of radiation incident upon the scatterer, the area of the x-ray beam at the scatterer, the distance between the scatterer and the location of interest, and the angle of scattering of the x rays. For a scattering angle of 90 degrees with respect to a diagnostic x-ray beam 400 cm² in cross-sectional area that is incident upon a scatterer, the exposure rate 1 m from the scatterer is less than 1/1000 of the primary beam exposure rate at the scatterer (Margin Figure 28-4). At other distances from the scatterer, the exposure rate varies as $1/d^2$.

Over a week, the exposure at the scatterer (the patient) is

$$\begin{aligned} \text{Exposure at scatterer} &= \frac{WUT}{d^2} \\ &= \frac{W}{d^2} \end{aligned}$$

where U and T are 1 because the beam is always directed at the patient and a patient is always present. At some location a distance d' from and at right angles to the primary beam, the exposure due to scattered radiation is

$$\begin{aligned} \text{Exposure per week from scattered radiation} &= \frac{(W/d^2)(1/1000)(F/400)}{(d')^2} \end{aligned} \tag{28-4}$$

where 1/1000 is the ratio of scattered radiation at 1 m to primary radiation at the scatterer, F is the area of the x-ray beam at the scatterer, $F/400$ corrects the ratio 1/1000 for more or less scattered radiation from fields larger or smaller than 400 cm², and $(d')^2$ corrects the scattered radiation for the inverse square falloff of exposure with distance. The use factor is always 1 for scattered radiation. With insertion of an occupancy factor T for the location of interest, the exposure per week to an individual from scattered radiation may be estimated as

$$\begin{aligned} \text{Exposure to an individual due to scattered radiation (R/wk)} &= \frac{(W/d^2)(1/1000)(F/400)(T)}{(d')^2} \end{aligned} \tag{28-5}$$

To reduce the exposure to an acceptable level X_p , a secondary barrier may be interposed between the scatterer and the location of interest. The barrier introduces an

attenuation factor B into Eq. (28-5):

$$X_p = \frac{(W/d^2)(1/1000)(F/400)(T)}{(d')^2}(B)$$

The attenuation factor B is

$$B = \frac{X_p(d')^2}{(W/d^2)(1/1000)(F/400)(T)} \quad (28-6)$$

In diagnostic radiology, x-ray beams are almost always generated at tube voltages less than 500 kVp. Radiation scattered from these beams is almost as energetic as the primary radiation, and the barrier thickness for scattered radiation is determined from curves for the primary radiation.

Example 28-3

A diagnostic x-ray generator has a busy workload of 3000 mA-min/wk at 100 kVp and an average field size of 20×20 cm. The x-ray tube is positioned 1.5 m from a wall between the radiation room and a urology laboratory with uncontrolled access. The wall contains 3 cm of ordinary concrete. For an occupancy factor of 1, what thickness of lead should be added to the wall if the primary beam is not directed toward the wall?

$$B = \frac{X_p(d')^2}{(W/d^2)(1/1000)(F/400)(T)} \quad (28-6)$$

If the radiation scattered to the wall originates within a patient positioned 1 m below the x-ray tube, then

$$B = \frac{(0.01)(1.5)^2}{[3000/(1)^2](1/1000)(400/400)(1)} = 0.0075$$

where the maximum permissible exposure is 0.01 R/wk for persons in the urology laboratory. From the 100-kVp curve in Margin Figure 28-3, the thickness of lead required is 0.8 mm for an attenuation of 0.0075 at 100 kVp.

Secondary Barriers for Leakage Radiation

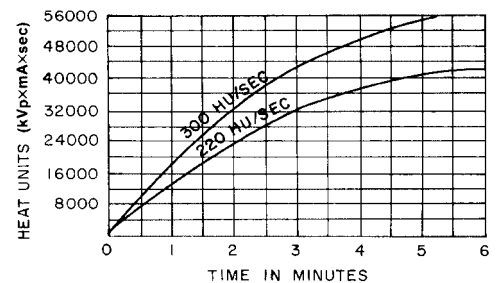
Radiation escaping in undesired directions through the x-ray tube housing is termed *leakage radiation*. According to the ICRP, this radiation should not exceed 0.1 R/hr at 1 m at the highest rated kilovolt-peak for the x-ray tube and at the highest milliamperage permitted for continuous operation of the x-ray tube. To determine the maximum tube current for continuous operation, the anode thermal characteristics chart for the x-ray tube must be consulted (Margin Figure 28-5). From this chart, it is possible to determine the maximum heat units* per second that the x-ray tube can tolerate during continuous operation. By dividing this value by the maximum rated kilovolts peak for the x-ray tube, the maximum milliamperage for continuous operation is computed. At this milliamperage value, the exposure rate due to leakage radiation should not exceed 0.1 R/hr at 1 m from the x-ray tube.

To determine the maximum exposure per week at 1 m due to leakage radiation, the workload in mA-min/wk is divided by the maximum milliamperage for continuous operation. The quotient is the effective operating time W' of the x-ray unit for purposes of computing the exposure due to leakage radiation. The exposure at a location of interest a distance d from the x-ray tube is

$$\text{Exposure due to leakage radiation} = \frac{(0.1)(W')}{60d^2}$$

*Heat units are discussed in Chapter 5.

William Herbert Rollins, a Boston dentist, was the first to suggest the use of the intensifying screen in 1902 as a means of reducing patient dose. Such screens were not commonly used until the 1940s. Among other contributions that Rollins made to radiation safety practices were lead shielding, collimators, and the use of increased kVp to decrease patient skin dose.



MARGIN FIGURE 28-5

A typical anode thermal characteristics chart.

X rays were discovered in November 1895. By March 1896 there were already reports of adverse symptoms related to x-ray exposure. In late 1896, a physician purposely exposed the little finger of his left hand to x rays on repeated occasions over several days and wrote about the pain, swelling, and stiffness that resulted. The episode that finally convinced most experts that x rays are associated with adverse bioeffects was the death in 1902 of Clarence Dally, one of Thomas Edison’s assistants. Dally’s job was to energize x-ray tubes as they were finished to determine if their output was sufficient for imaging. The pattern of skin cancer matched the parts of his body that were habitually close to the x ray tubes. After this incident, Edison ceased all research involving x rays.

Source: Brown, P., *American Martyrs to Science Through the Roentgen Rays*. Charles C Thomas, Springfield, IL, 1936.

where W' has units of minutes per week and 60 is a conversion from minutes to hours. The use factor for leakage radiation is always 1 because leakage radiation is emitted in all directions whenever the x-ray tube is used. To determine the exposure to an individual due to leakage radiation, an occupancy factor T for the location of interest should be added to the equation:

Exposure to individual from leakage radiation = $\frac{0.1W'T}{60d^2}$ (28-7)

By adding an attenuation factor B to the equation, leakage radiation may be reduced to the permissible level X_p :

$$X_p = \frac{0.1W'T}{60d^2} (B)$$

The attenuation factor B is

$$B = \frac{60X_p d^2}{0.1W'T} \tag{28-8}$$

Because leakage radiation is at least as energetic as the primary beam, the thickness of a barrier that provides the attenuation factor B is determined from curves in Margin Figure 28-3 for the primary x-ray beam.

A barrier designed for primary radiation provides adequate protection against scattered and leakage radiation. For a barrier designed for scattered and leakage radiation only (i.e., a secondary barrier), the thickness is computed separately for each type of radiation. If the barrier thickness required for scattered radiation differs by more than three half-value layers from that required for leakage, the greater thickness is used for the barrier. If the thicknesses computed for scattered and leakage radiation differ by less than three half-value layers, then the thickness of the barrier is increased by adding one half-value layer to the greater of the two computed thicknesses. Half-value layers for diagnostic x-ray beams are listed in Table 28-8.¹¹

Example 28-4

For the x-ray tube with a maximum rated kilovolt-peak of 100 and the anode thermal characteristics chart shown in Margin Figure 28-5, determine the maximum milliamperage for continuous operation and the thickness of lead that should be added

TABLE 28-8 Half-Value Layers in Millimeters of Lead and in Centimeters and Inches of Concrete for X-Ray Beams Generated at Different Tube Voltages

Attenuating Material	Half-Value Layer						
	50 kVp	70 kVp	100 kVp	125 kVp	150 kVp	200 kVp	250 kVp
Lead (mm)	0.05	0.15	0.24	0.27	0.29	0.98	0.9
Concrete (in.)	0.17	0.33	0.6	0.8	0.88	1.0	1.1
Concrete (cm)	0.43	0.84	1.5	2.0	2.2	2.5	2.8
Attenuating Material	Half-Value Layer						
	300 kVp	400 kVp	500 kVp	1,000 kVp	2,000 kVp	3,000 kVp	
Lead (mm)	1.4	2.2	3.6	7.9	12.7	14.7	
Concrete (in.)	1.23	1.3	1.4	1.75	2.5	2.9	
Concrete (cm)	3.1	3.3	3.6	4.4	6.4	7.4	

Source: National Council on Radiation Protection and Measurements. *Medical X-Ray and Gamma-Ray Protection for Energies Up to 10 MeV*. Recommendations of the NCRP, Report No. 34, Washington, D.C., NCRP, 1970.

to the wall described in Example 28-3 to protect against both scattered and leakage radiation.

From Margin Figure 28-6, the maximum anode thermal load that can be tolerated by the x-ray tube is 300 heat units (HU) per second.

$$\text{Maximum mA for continuous operation of maximum kVp} = \frac{300 \text{ HU/sec}}{100 \text{ kVp}} = 3 \text{ mA} \quad (28-8)$$

$$\text{Effective operating time } W' = \frac{3000 \text{ mA-min/wk}}{3 \text{ mA}} = 1000 \text{ min/wk}$$

$$\begin{aligned} B &= \frac{60X_p d^2}{0.1W'T} \\ &= \frac{60(0.01)(1.5)^2}{0.1(1000)(1)} \\ &= 0.0135 \end{aligned}$$

From Margin Figure 28-3, the thickness of lead required to protect against leakage radiation is 0.6 mm Pb. From Example 28-3, the thickness of lead required to protect against scattered radiation is 0.8 mm Pb. The difference between these thicknesses is less than three times the half-value layer of 0.24 mm Pb for a 100-kVp x-ray beam (see Table 28-8). Hence, the total thickness of lead required to shield against both scattered and leakage radiation is $(0.8 + 0.24) = 1.04$ mm Pb. The concrete in the wall provides shielding equivalent to 0.4 mm Pb, so 0.64 mm, or 1/32 in., of lead must be added to the wall.

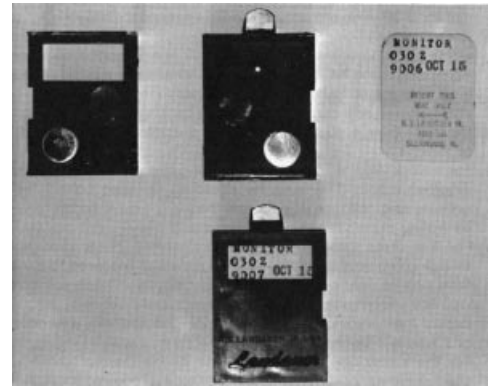
Special Shielding Requirements for X-Ray Film

X-ray film is especially sensitive to radiation, and film storage areas and bins must often be shielded to a greater extent than areas occupied by individuals. For film storage areas, shielding computations are identical with those described above, except that maximum permissible exposures X_p must be reduced to much lower values. As a rule of thumb, the exposure should be limited to no more than 1 mR for the maximum storage time of the film (e.g., 2 weeks).

Changes in Shielding Recommendations

At the present time, the National Council on Radiation Protection and Measurements is in the process of revising its 26-year-old recommendations on shielding design for diagnostic radiology. The changes are expected to include:

- Revisions of workload estimates that take into account the range of kVps that are currently used.
- Leakage and scatter calculations will be modeled more explicitly with regard to values found in modern equipment. The present rule of adding a half-value layer if the leakage and scatter barrier requirements are similar will be abandoned, and both sources of exposure will be modeled together.
- Currently, occupancy factors are not allowed to be lower than 1/16. The revised guidelines will allow for spaces such as closets, stairways, storage facilities, and so on, to be assigned occupancy factors as low as 1/40.
- Current guidelines do not take into account some attenuating material that is present. These materials include the image receptor assembly and the patient table. In the new guidelines, the attenuating properties of these materials will be considered.
- Use factors will be revised to reflect current practices.



MARGIN FIGURE 28-6

Film badge for measuring the exposure of individuals to x and γ radiation and to high-energy β radiation. (Courtesy of R. S. Landauer, Jr., and Co.)

Estimation of Patient Dose in Diagnostic Radiology

The radiation dose delivered to a patient during a radiographic procedure may be estimated with reasonable accuracy with the rule of thumb of 1 R/mA-min at 1 m from a diagnostic x-ray unit. Frequently, this estimate is needed when a patient is discovered shortly after an examination to have been in an early stage of pregnancy when the x-ray procedure was performed. In this case, an estimate of fetal dose is required. A simple procedure for obtaining a rough estimate of fetal dose is given in Example 28-5.

Example 28-5

A patient receives 2 minutes of fluoroscopy (80 kVp, 3 mA) and four spot films (80 kVp, 30 mA-s) over the 20-cm thick lower abdominal region. The x-ray tube-to-tabletop distance is 0.5 m for the x-ray unit used for the examination. A short time later it is determined that the patient was 3 weeks pregnant at the time of the examination. Estimate the dose to the fetus.

The rule of thumb of 1 R/mA-min yields $[1 \text{ R/mA-min}/(0.5)^2]$, or 4 R/mA-min, at a distance of 0.5 m. The exposure at the skin delivered during fluoroscopy is

$$\begin{aligned}\text{Fluoroscopic exposure} &= \left(4 \frac{\text{R}}{\text{mA-min}}\right) (3 \text{ mA})(2 \text{ min}) \\ &= 24 \text{ R}\end{aligned}$$

The exposure at the skin delivered by the spot films is

$$\begin{aligned}\text{Spot film exposure} &= \frac{(4 \text{ R/mA-min})(30 \text{ mA-sec/film})(4 \text{ films})}{60 \text{ sec/min}} \\ &= 8 \text{ R}\end{aligned}$$

The total exposure at the skin is $24 \text{ R} + 8 \text{ R} = 32 \text{ R}$. For average-size patients, the transmission of the x-ray beam through the abdomen is roughly 0.1% to 0.5% of the incident exposure. A transmission of 0.4% means that the patient provides 8 HVL's of attenuation. With the assumption that the fetus is positioned midway in the abdomen, shielding equivalent to 4 HVL's (6.2% transmission) is provided to the fetus. With a conservative estimate of 7%, the exposure to the fetus is

$$\begin{aligned}\text{Exposure to fetus} &= (32 \text{ R})(0.07) \\ &= 2.2 \text{ R}\end{aligned}$$

The absorbed dose in milligray is approximately equal numerically to ten times the exposure in roentgens, so the absorbed dose to the fetus is roughly 22 mGy (2.2 rad).

The average glandular dose during mammography is 2.5–3.0 mSv for a 4.5-cm compressed breast composed of 50% fat and 50% glandular tissue, two-view study (craniocaudal and oblique).

Source: McLelland, R. et al. *AJR* 1991; 157:473–479.

Values of entrance surface absorbed dose (skin dose, ESAD) are given in Table 28-9, and multiplicative factors that may be used to convert ESAD to fetal dose are given in Table 28-10. Typical bone marrow doses for radiographic exams are given in Table 28-11. For actual patient and fetal dosimetry, a qualified medical physicist can determine a more accurate estimate of dose based upon specific output measurements from the source and data such as patient thickness, technique factors, and so on.

■ AREA AND PERSONNEL MONITORING

The radiation exposure must be known for persons working with or near sources of ionizing radiation. This knowledge may be obtained by periodic measurements of exposure rates at locations accessible to the individuals. Furthermore, persons working with or near sources of ionizing radiation should be equipped with personnel monitors

TABLE 28-9 Mean Entrance Surface Absorbed Dose (ESAD) to Soft Tissue

Examination (Projection) ^a	ESAD
Film/screen	
Chest (P/A)	0.20 mGy
Skull (lateral)	1.93 mGy
Abdomen (kidneys, ureters, bladder) (A/P)	6.27 mGy
Retrograde pyelogram (A/P)	4.98 mGy
Thoracic spine (A/P)	2.85 mGy
Lumbar spine (A/P)	6.41 mGy
Full spine (A/P)	2.03 mGy
Feet (D/P)	0.70 mGy
Non-film/screen	
Fluoroscopy	<100 mGy min ⁻¹
Cineradiography ^b	250 mGy min ⁻¹
Digital radiography	2.5 mGy/image
Computed tomography	10–50 mGy

^aP/A refers to posterior (back) to anterior (front) beam projection (i.e., the x-ray beam enters the posterior portion of the body and exits anteriorly). A/P refers to anterior to posterior beam projection. D/P refers to dorsal (top) to pedal (bottom of foot) beam projection.

^b0.14 mGy/frame, 30 frame s⁻¹.

Source: Ritenour, E. R., Geise, R. A., Radiation Sources: Medicine, in Hendee, W. R., and Edwards, F. M. (eds.), *Health Effects of Exposure to Low-Level Ionizing Radiation*. Philadelphia, Institute of Physics Publishing, 1996, p. 440.

that reveal the amount of exposure the individuals have received. The monitor used most frequently is the film badge containing two or more small photographic films enclosed within a light-tight envelope. The badge is worn for a selected interval of time (1 to 4 weeks) at some convenient location on the clothing over the trunk of the body. During fluoroscopy, most persons wear the film badge under a lead apron to monitor the exposure to the trunk of the body, although some state regulations require a second badge on or near the collar to measure the dose to organs (e.g., thyroid gland) not shielded by the apron. After the desired interval of time has elapsed, the film is processed, and its optical density is compared with the optical density of similar films that have received known exposures. From this comparison, the radiation exposure may be determined for the film badge and, supposedly, for the individual wearing the film badge. Small metal filters mounted in the plastic holder for the film permit some differentiation between different types of radiation and different energies of x rays that contribute to the exposure (Margin Figure 28-6). Types of radiation that may be monitored with a film badge include x and γ rays, high-energy electrons, and neutrons. Special holders have been designed for wrist badges and ring badges that are used to estimate the exposure to limited regions of the body. Although film badges furnish a convenient method of personnel monitoring, the difficulties of accurate dosimetry with photographic film limit the accuracy of measured exposures. Other methods for personnel dosimetry include pocket ionization chambers and thermoluminescent and photoluminescent dosimeters.

Personnel monitors such as badges do not provide a complete dosimetric picture of individual exposure. However, effective dose may be estimated from them. They are usually calibrated to estimate dose at a depth of 10 mm in soft tissue. This concept is referred to in federal radiation safety regulations as *deep dose equivalent*.

Calculation of effective dose from badge readings is of particular concern for personnel who perform fluoroscopy on a regular basis and who may approach equivalent dose limits. An extensive review of the relationships among effective dose equivalent, x-ray tube output, apron thickness, and geometric considerations in neurointerventional fluoroscopy environments¹² has yielded the following equation¹³:

$$E = 0.5H_W + 0.025H_N \quad (28-9)$$

TABLE 28-10 Typical Fetal Dose Factors as a Function of Entrance Surface Absorbed Dose (ESAD)^a

X-Ray Exam	Fetal Dose Factor mGy/[mGy ^{ESAD}]
Skull	<0.01
Cervical	<0.01
Limb	<0.01
Chest	0.03
Stomach and upper GI	0.30
Lumbar spine	2.90
Abdomen	3.05
Pelvis	3.40

^aCalculated from data found in Ballinger, P. W. Merrill's *Atlas of Radiographic Positions and Radiologic Procedures*, Vol. 1, 8th edition, St Louis, Mosby, 1995.

TABLE 28-11 Typical Bone Marrow Doses for Various Radiographic Examinations

X-Ray Examination	Mean Marrow Dose (mrad)
Skull	10
Cervical	20
Chest	2
Stomach and upper gastrointestinal	100
Gallbladder	80
Lumbar spine	60
Intravenous urography	25
Abdomen	30
Pelvis	20
Extremity	2

Source: Ballinger, P. W. Merrill's *Atlas of Radiographic Positions and Radiologic Procedures*, Vol. 1, 8th edition, St Louis, Mosby, 1995.

where E is effective dose, H_W is the deep dose equivalent obtained from a badge worn at the waist under an apron, and H_N is the deep dose equivalent obtained from a badge worn at the neck outside the apron. If a badge is worn at the neck only, the effective dose may be estimated from

$$E = 0.048H_N \quad (28-10)$$

Equations (28-9) and (28-10) are intended to provide an estimate of effective dose that is two to three times the actual effective dose received by the badged worker.

PROBLEMS

- *28-1. What is the maximum permissible accumulated effective dose equivalent for a 30-year-old occupationally exposed individual?
- *28-2. A nuclear medicine technologist received an estimated 200 mrem whole-body dose equivalent during the first 2 months of pregnancy. According to the NCRP, what is the maximum permissible dose equivalent over the remaining 7 months of pregnancy?
- *28-3. The exposure rate constant Γ_∞ is 0.22 R-m²/hr-Ci for ¹³¹I. What is the exposure over a 40-hour week at a distance of 2 m from 200 mCi of ¹³¹I?
- *28-4. For a workload of 750 mA-min/wk for a dedicated 125-kVp chest radiographic unit, determine the shielding required behind the chest cassette at a distance of 6 ft from the x-ray tube if an office with uncontrolled access is behind the cassette.
- *28-5. For the chest radiographic unit in Problem 28-4, determine the shielding required to protect against radiation scattered to a wall 2 m from and at right angles to the chest cassette if the area behind the wall has uncontrolled access with an occupancy of one.
- *28-6. Repeat the computation in Problem 28-5 if film with a turnover time of 2 weeks is stored behind the wall.
- *28-7. For the x-ray tube described by the anode thermal characteristics chart in Margin Figure 28-5, determine the shielding required for leakage radiation at a wall 2 m from the x-ray tube if the area behind the wall has uncontrolled access with an occupancy of one. The x-ray tube is rated at 125 kVp maximum.
- *28-8. For the wall in Problem 28-7, 1.0 mm of Pb is required to shield against scattered radiation. What thickness of shielding is required to protect against both scattered and radiation leakage?

*For those problems marked with an asterisk, answers are provided on p. 493.

SUMMARY

- The dose limits recommended by the NCRP include:
 - Stochastic limits in terms of effective dose
 - Organ limits in terms of equivalent dose
- The United Nations Scientific Committee on the Effects of Atomic Radiation estimates that the lifetime cancer risk from radiation received during the fetal stage is two to three times that of the adult (i.e., approximately 10% per Sv).
- After a pregnancy is declared by a worker, an equivalent dose limit of 0.5 mSv per month is recommended for the fetus.
- An estimate of occupational effective dose E can be derived from monitor badge deep dose equivalent readings at the waist, H_W , and at the neck, H_N by the following equation:

$$E = 0.5H_W + 0.025H_N$$

If only the neck badge is worn, then the estimate is

$$E = 0.048H_N$$

- Groups that provide recommendations and information on the subject of radiation protection include:
 - National Council on Radiation Protection and Measurements (NCRP)
 - International Commission on Radiological Protection (ICRP)

- United Nations Scientific Committee on the Effects of Atomic Radiation (UNSCEAR)
- National Academy of Sciences (NAS)
- Conference of Radiation Control Program Directors (CRCPD)
- Groups that have legal authority to regulate manufacture and use of sources of ionizing radiation include:
 - Nuclear Regulatory Commission (NRC)
 - State Health Departments
 - Environmental Protection Agency (EPA)
 - Food and Drug Administration (FDA)
- The **collective effective dose** is defined as the average dose estimated for individuals in the population times the total number of individuals in the population. It is used to predict the magnitude of stochastic effects in a population.
- The **genetically significant dose** is an estimate of the “genetic impact” of radiation exposures to a population. It is based upon estimates of the fraction of the population that undergo diagnostic and nuclear medicine exams each year, in addition to all other sources of background exposure. It is weighted by the age and sex distribution of the population to take into account the expected number of children that would be produced by each category during the remainder of their lifetimes.
- The NCRP has based its process of recommending limits on radiation exposure with the goal of maintaining the risk of genetic effects, fatal cancers, and nonfatal cancers at the same level as that found in “safe” industries. Their estimates show that the lifetime risk from an average worker’s exposure values over a working career is approximately 1 in 10,000. Maximum exposure levels would result in a lifetime risk of 5 in ten thousand.
- The attenuation factor that must be provided by shielding material is calculated from knowledge of workload W , use factor U , occupancy factor T , and distance- d as follows:

$$B = \frac{X_p d^2}{WUT}$$

REFERENCES

1. Claus, W. The Concept and Philosophy of Permissible Dose to Radiation, in Claus, W. (ed.), *Radiation Biology and Medicine*. Reading, MA, Addison-Wesley, 1958, p. 389.
2. National Council on Radiation Protection and Measurements: *Limitation of Exposure to Ionizing Radiation*, Report No. 116. Bethesda, Maryland, National Council on Radiation Protection and Measurements, 1993.
3. International Commission on Radiological Protection: *1990 Recommendations of the International Commission on Radiological Protection*, ICRP Publication No. 60, Annals of the ICRP 21. Elmsford, New York, Pergamon Press, 1991.
4. Food and Drug Administration: *Recommendations for Evaluation of Radiation Exposure from Diagnostic Radiology Examinations*, HHS Publication (FDA) 85-8247. Springfield, VA, National Technical Information Service, 1985.
5. Environmental Protection Agency. Radiation protection guidance to federal agencies for diagnostic x rays. *Fed. Reg.* 1978; **43**(22):4377.
6. National Council on Radiation Protection and Measurements: *Limitation of Exposure to Ionizing Radiation*, Report No. 116. Bethesda, Maryland, National Council on Radiation Protection and Measurements, 1993, p. 36.
7. United Nations Scientific Committee on the Effects of Atomic Radiation. *Genetic and Somatic Effects of Ionizing Radiation*, Report to the General Assembly with Annexes. United Nations Publications, New York, 1986.
8. International Commission on Radiological Protection. *Cost-Benefit Analysis in the Optimization of Radiation Protection*, ICRP Publication No. 37. New York, Pergamon Press, 1977.
9. National Council on Radiation Protection: *Medical X-Ray, Electron Beam and Gamma-Ray Protection for Energies up to 50 MeV (Equipment Design, Performance and Use)*. Recommendations of the NCRP, Report No. 102. Washington, D.C., 1989.
10. Appleton, G., and Krishnamoorthy, P. *Safe Handling of Radioisotopes: Health Physics Addendum*. Vienna, International Atomic Energy Agency, 1960.
11. *Structural Shielding Design and Evaluation for Medical Use of X Rays and Gamma Rays of Energies Up to 10 MeV*. Recommendations of the NCRP, Report No. 49. Washington, D.C., 1976.
12. Faulkner, K., and Marshall, N. W. The relationship of effective dose to personnel and monitor reading for simulated fluoroscopic irradiation conditions. *Health Phys.* 1993; **64**:502–508.
13. Webster, E. W. EDE for exposure with protective aprons. *Health Phys.* 1989; **56**:568–569.

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■ OBJECTIVES

After studying this chapter, the reader should be able to:

- Define the expressions committed dose equivalent, annual limit on intake, and derived air concentration.
- Distinguish between effective half-life for uptake and effective half-life for elimination, and identify the influencing variables on each.
- Identify the factors that influence the dose rate to an organ from an internally deposited radionuclide.
- Define the absorbed fraction, the absorbed dose constant, cumulative activity, and mean absorbed dose per cumulative activity.
- Explain the concept of “locally absorbed radiation” and the types of radiation it encompasses.
- Perform a few simple internal dose computations.

■ INTRODUCTION

Hazards associated with internal sources of radiation are best controlled by minimizing the absorption, inhalation, and ingestion of radioactive materials. Protection standards for internal sources of radiation are based on the concept of *committed dose equivalent* (CDE).

■ COMMITTED DOSE EQUIVALENT

The radiation dose to an organ from a radioactive nuclide deposited in the body is delivered over a period of time as the nuclide is eliminated from the body. The time required for the nuclide to be eliminated by biological processes or radioactive decay depends on the physical characteristics of the nuclide and its chemical form. To cover a wide range of radiopharmaceuticals the International Commission on Radiological Protection (ICRP) has defined¹ the *committed dose equivalent* (CDE) for an organ as the dose accumulated in the organ over a 50-year period. The CDE is obtained by integrating (summing) the dose equivalent rate over time. The dose equivalent rate in an organ is determined from knowledge of the spatial distribution of the nuclide, the type of radiation emitted, and the range of the radiation in tissue. Finally, the committed effective dose equivalent (CEDE) (see Chapter 28) may be calculated by multiplying the CDE by the weighting factor for the organ. The committed effective dose equivalent for the entire body is determined by summing the CEDE for all organs containing the radioactive nuclide.

To prevent workers from accumulating internal nuclides that would exceed an acceptable CDE, limits have been set for the amount of various nuclides that may be taken into the body during any 1 year. These annual limits on intake (ALI) are based upon mathematical models for retention and excretion of radionuclides from the adult human body.^{2,3} Some examples of ALIs are given in the margin.

The workplace where radioactive materials are handled must be designed so that workers do not exceed the intake and dose limits. It is not practical to monitor the actual uptake in workers continuously for all nuclides that may be present in the workplace. It is also not consistent with the principles of as low as reasonably achievable (ALARA) to base safety procedures solely upon a response to accidental exposures. It is more appropriate to design the workplace so that concentrations of potential contaminants are kept well below levels that could lead to values near the ALI. To aid in this approach to radiation protection, the derived air concentration (DAC) is defined as the in-air concentration of a radionuclide that would

Radioactive substances can enter the body accidentally by mechanisms of absorption, inhalation and ingestion.

The *residence time* of a radioactive nuclide in an organ is the time that would be required to deliver the radiation dose to an organ if the initial activity in the organ remained constant until it had been completely eliminated, at which time it would decrease precipitously to zero.

Annual Limits on Intake for Selected Radionuclides

Radionuclide	GBq (yr)
Hydrogen 3	1.8
Carbon 14	0.088
Phosphorus 32	0.036
Technetium 99m	5.3
Iodine 131	0.0018
Xenon 133	8.8